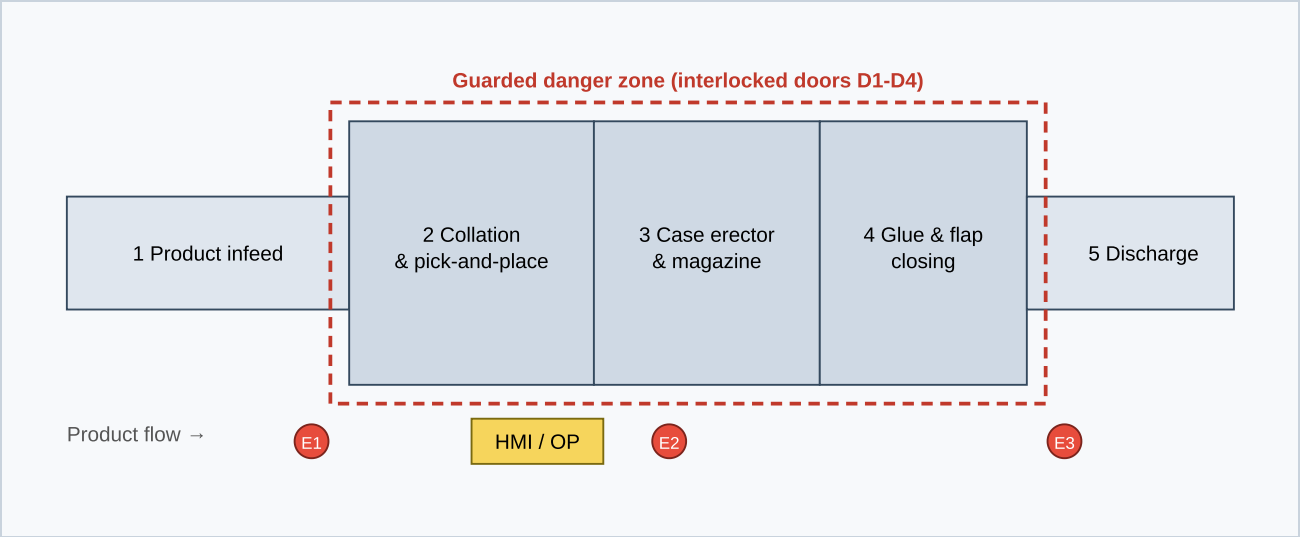
FICTIONAL SAMPLE — example data

EXAMPLE MACHINERY CO. (FICTIONAL)

SAMPLE-A Case Packer

Operating and Maintenance Manual



Model	SAMPLE-A
Serial number	_____
Document	OM-SAMPLE-A-EN
Revision	D · October 2026
Language	English (original instructions)

⚠ Read and understand this manual before installing, operating or servicing the machine. Keep it with the machine.

FICTIONAL SAMPLE created to show ManualOS output. The machine, company and all data are invented. Do not use for real equipment.

Contents

0 About This Manual 3
- 0.1 Revision History 3
- 0.2 Purpose and Scope 3
- 0.3 Who Should Read Which Chapters 3
- 0.4 Safety Messages, Symbols and Conventions 3
- 0.5 Related Documents 4

1 Safety 5
- 1.1 General Safety Instructions 5
- 1.2 Personnel Qualifications and Responsibilities 5
- 1.3 Intended Use 5
- 1.4 Reasonably Foreseeable Misuse 6
- 1.5 Danger Zones and Operator Positions 6
- 1.6 Safety Devices and Safety Functions 7
- 1.7 Emergency Stop 7
- 1.8 Safety Signs and Labels 8
- 1.9 Personal Protective Equipment 8
- 1.10 Energy Isolation and Lockout/Tagout 8
- 1.11 Residual Risks 10
- 1.12 Noise, Vibration and Emissions 11
- 1.13 Fire Prevention 11
- 1.14 Cybersecurity 12

2 Machine Description and Theory of Operation 13
- 2.1 What the Machine Does 13
- 2.2 Machine Layout and Product Flow 13
- 2.3 Main Assemblies and Their Functions 13
- 2.4 How the Infeed Conveyor Works 14
- 2.5 How the Collation and Pick-and-Place Works 14
- 2.6 How the Case Erector Works 16
- 2.7 How the Glue and Flap Closing Works 16
- 2.8 How the Case Discharge Works 17
- 2.9 How the Air and Vacuum System Works 18
- 2.10 How the Electrical and Control System Works 18
- 2.11 Sequence of Operation 19
- 2.12 Interlocks, Permissives and Machine States 20
- 2.13 Control Logic Description 21
- 2.14 Identification and Nameplate 23
- 2.15 Options and Accessories 24

3 Transport, Handling and Storage 24
- 3.1 Transport Dimensions and Masses 24
- 3.2 Lifting 24
- 3.3 Unpacking and Inspection on Receipt 25
- 3.4 Storage 25

4 Installation 25
- 4.1 Site Requirements and Foundation 25
- 4.2 Mechanical Installation and Leveling 26
- 4.3 Electrical Supply Connection 26
- 4.4 Compressed Air Connection 27
- 4.5 Control and Network Connections 27
- 4.6 Noise and Vibration Measures 27

5 Commissioning 28
- 5.1 Pre-Start Checklist 28
- 5.2 Initial Start-Up and Test Run 28
- 5.3 Safety Function Verification 29
- 5.4 Re-commissioning After Standstill or Relocation 29
- 5.5 Operator Training and Handover 29

6 Controls and Operator Panel 30
- 6.1 Operator Panel Layout 30
- 6.2 Stack Light 30
- 6.3 User Levels and Login 30
- 6.4 Screen Navigation 31
- 6.5 Home Screen 31
- 6.6 Recipes Screen 32
- 6.7 Manual Control Screen 33
- 6.8 Alarms Screen 33
- 6.9 Maintenance Screen 34

7 Operation 35
- 7.1 Operating Modes 35
- 7.2 Daily Start-Up 35
- 7.3 Normal Production 35
- 7.4 Format Changeover 36
- 7.5 Recipe Parameters 37
- 7.6 Stopping and Shutdown 37
- 7.7 Cleaning 38

8 Alarms and Troubleshooting 39
- 8.1 Alarm Messages 39
- 8.2 Troubleshooting Guide 40
- 8.3 Clearing Jams 41
- 8.4 Emergencies and Rescue 42

9 Maintenance 43
- 9.1 Maintenance Safety 43
- 9.2 Preventive Maintenance Schedule 43
- 9.3 Lubrication 44
- 9.4 Inspection and Safety-Function Testing 45
- 9.5 Checking and Adjusting Belt Tension 45
- 9.6 Cleaning the Glue System 46
- 9.7 Replacing Vacuum Cups 47
- 9.8 Aligning Sensors 47
- 9.9 Adjustment and Calibration 48

10 Spare Parts 48
- 10.1 Recommended Spare Parts 48
- 10.2 Ordering Parts and Service 49

11 Decommissioning and Disposal 49
- 11.1 Decommissioning and Dismantling 49
- 11.2 Disposal 50

12 Appendices 51
- 12.1 Technical Data 51
- 12.2 Electrical Overview 52
- 12.3 Pneumatic Overview 53
- 12.4 I/O List 53
- 12.5 Control Program Tag List 55
- 12.6 Alarm List with Stop Reactions 55
- 12.7 Control Program Cross-Reference 56
- 12.8 Tightening Torques 57
- 12.9 Lubricants and Consumables 58
- 12.10 Maintenance Log 58
- 12.11 Warranty and Service Contacts 58
- 12.12 Glossary 59
- 12.13 Detailed Revision History 59
- 12.14 Index 60

0 About This Manual

0.1 Revision History

Revision	Date	Description	Approved
A	2026-03-02	First release	J. Ortega
B	2026-07-15	Added low-glue alarm and 2,000-hour service tasks	J. Ortega
C	2026-09-01	Updated spare parts list; new operator panel screens	M. Patel
D	2026-10-24	Complete revision: expanded safety, installation, commissioning, procedures, troubleshooting and appendices	M. Patel

A detailed list of changes in each revision is given in Appendix 12.13.

0.2 Purpose and Scope

This manual explains how to transport, install, commission, operate, maintain and decommission the SAMPLE-A Case Packer safely. It applies to all SAMPLE-A machines from serial number SA-0101 with control software version 3.x. Keep the manual next to the machine for its entire service life and pass it on if the machine is sold.

The manual is the original instruction in English. It does not replace site safety rules, operator training, or the separate instructions for bought-in components such as the hot-melt unit.

0.3 Who Should Read Which Chapters

Reader	Read before starting work	Keep for reference
Operator	Chapters 1, 2, 6 and 7	Chapter 8 (alarms), daily tasks in Chapter 9
Setup technician	Chapters 1, 2, 6 and 7, Section 7.4	Chapters 8 and 9
Mechanical maintenance	Chapters 1, 2, 8 and 9	Chapter 10, Appendices
Electrical maintenance	Chapters 1, 2, 4, 8 and 9	Appendices 12.2 to 12.7
Installer / commissioning engineer	Chapters 1, 3, 4 and 5	Appendix 12.1

0.4 Safety Messages, Symbols and Conventions

Safety messages in this manual follow ANSI Z535.6 and ISO 3864-2. Each message names the hazard, explains what can happen, and tells you how to avoid it. In procedures the message is placed directly **before** the step it applies to.

Signal word	Meaning
⚠ DANGER	A hazardous situation that **will** result in death or serious injury if not avoided.
⚠ WARNING	A hazardous situation that **could** result in death or serious injury if not avoided.
⚠ CAUTION	A hazardous situation that could result in minor or moderate injury if not avoided.
NOTICE	Information about practices that could damage the machine or property. Not related to personal injury.

Typographic conventions

Convention	Meaning
Start	A button on the operator panel or a touch key on the screen
Recipes	A screen on the operator panel
Numbered steps	Do the steps in order; each step has one action
→ Expected result	What you should see when the step is done correctly. If you do not, stop and see Chapter 8
Inferred – please confirm *Example.*	Information derived from the machine data that the manufacturer must confirm

0.5 Related Documents

Document	Number	Supplied as
Electrical schematics	E-SA rev. D	PDF and paper in cabinet pocket
Pneumatic schematic	P-SA rev. C	PDF and paper
Illustrated parts catalog	IPC-SA	PDF
Hot-melt unit manual	Supplier document	Paper, with the unit
Declaration of conformity	DoC-SAMPLE-A	Paper
Commissioning report	CR-SA-xxxx	Completed at site

1 Safety

1.1 General Safety Instructions

- Read this manual completely before working on the machine.
- Only trained and authorized persons may operate or service the machine (Section 1.2).
- Never operate the machine with a guard open, removed or damaged.
- Do not modify the machine or its control program without written approval from the manufacturer.
- Report any damage, unusual noise or malfunction to your supervisor immediately.
- Keep the area around the machine clean, dry and free of loose carton material.
- Follow all site safety rules in addition to this manual.

⚠ DANGER

Bypassing, removing or defeating interlocks, guards or safety functions.

Exposes people to all machine hazards and will lead to death or serious injury.

- Never bridge, remove or tamper with safety devices.
- Report defective safety devices and take the machine out of service until repaired.

1.2 Personnel Qualifications and Responsibilities

Role	Qualification	Permitted tasks
Operator	Trained on this machine (Section 5.5)	Start, stop, load case blanks, refill glue, clear jams outside the guard, daily checks
Setup technician	Operator training plus changeover training	Format changeover, recipe selection, adjustments in Setup mode
Mechanical maintenance	Qualified mechanic with lockout training	Mechanical maintenance, part replacement, lubrication, belt and sensor adjustment
Electrical maintenance	Qualified electrical worker (NFPA 70E)	Work in the electrical cabinet, drive and controller replacement, safety-function tests
Engineer (manufacturer or authorized)	Trained by the manufacturer	Program changes, safety configuration, recipe limits

The owner (operating company) is responsible for: making this manual available at the machine; training and authorizing personnel and keeping records; providing the PPE in Section 1.9; having the periodic safety tests in Section 9.4 carried out; and making sure the machine is used only as intended.

1.3 Intended Use

The SAMPLE-A is designed to erect corrugated regular slotted cases (RSC), load them with sealed packaged products (cartons, bottles or pouches) and close them with hot-melt adhesive. It is intended for indoor industrial use within the limits in Appendix 12.1.

- Products: 50–1,500 g per unit, maximum 12 kg per case
- Case sizes: length 200–450 mm, width 150–350 mm, height 100–350 mm
- Speed: up to 20 cases per minute
- Environment: 5–40 °C, 20–85 % relative humidity, non-condensing, non-explosive atmosphere

1.4 Reasonably Foreseeable Misuse

The following uses are prohibited:

- Packing unsealed food, loose liquids, or hazardous goods
- Operating with guard doors open or interlocks bypassed
- Reaching through the infeed or discharge tunnel openings
- Climbing on or riding the machine or its conveyors
- Using adhesives other than those in Appendix 12.9
- Operation in potentially explosive atmospheres
- Using the machine as a work platform or for storage

1.5 Danger Zones and Operator Positions

The danger zone is the fully guarded area from the end of the infeed conveyor to the case discharge. During production the operator works only at the operator panel and at the blank-magazine loading position, both outside the guard. Figure 1-1 shows the zones, the safety devices and the energy isolation points.

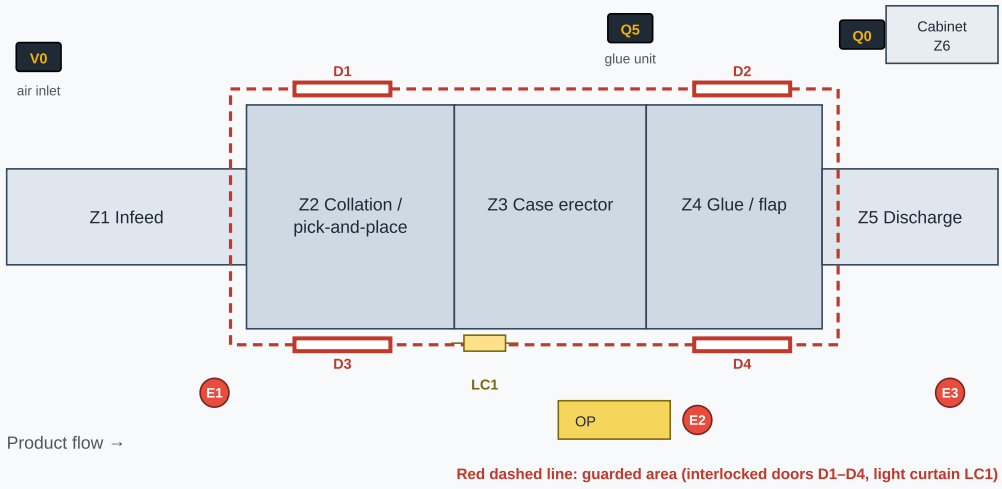


Figure 1-1. Danger zones Z1–Z6, guard doors D1–D4, light curtain LC1, emergency stops E1–E3, operator panel (OP) and isolation points Q0, Q5 and V0.

Zone	Area	Main hazards	Access during production
Z1	Infeed conveyor	Entanglement at rollers, drawing-in at metering transfer	Outside the guard only; side guides fixed
Z2	Collation and pick-and-place	Crushing under the head, impact from gantry, falling load	None – interlocked doors D1, D3
Z3	Case erector and magazine	Crushing by swing arm and folders	Magazine loading only, protected by LC1
Z4	Glue and flap closing	Burns from hot-melt, drawing-in at compression belts	None – interlocked doors D2, D4
Z5	Discharge conveyor	Drawing-in between rollers	Outside the guard only
Z6	Electrical cabinet	Electric shock, arc flash	Qualified electrical workers only

1.6 Safety Devices and Safety Functions

The safety functions are carried out by a certified safety controller and safety-rated drives. They are designed to ISO 13849-1 at the performance level shown. Never bypass, remove or adjust a safety device.

Device	Location	What it does	Stop	PL	Test interval
Emergency stop E1–E3	Infeed, operator panel, discharge	Stops all motion, removes drive torque and exhausts air	Cat. 1	d	Daily
Guard door interlocks D1–D4	All four access doors (coded, with guard locking on D1/D2)	Opening a door stops motion; restart only after closing and reset	Cat. 1	d	Daily
Light curtain LC1	Blank magazine	Stops the erector when the operator reaches into the magazine	Cat. 1	c	Weekly
Safe Torque Off (STO)	All drives	Removes motor torque on E-stop, door open or light curtain	Cat. 0/1	d	Yearly
Vertical axis brake	Pick-and-place head	Holds the head when the axis is not enabled; tested at every homing	—	c	Monthly
Dump valve Y01	Air inlet module	Exhausts machine air on E-stop and at lockout	—	c	Weekly
Enabling device	Pendant in Setup mode	Allows jog only while held in the middle position	—	d	Monthly
Fixed guards	Around drives and belt ends	Prevent access; can be removed only with tools	—	—	Monthly visual

Stop categories

Category	Meaning on the SAMPLE-A
Cat. 0	Power is removed from the drives immediately (uncontrolled stop). Used for safety controller faults and STO feedback errors.
Cat. 1	Drives brake all axes to standstill under control, then torque is removed within 0.5 s. Used for E-stop, door and light curtain.
Cycle stop	The current case is finished and all stations stop at home. Not a safety function.

1.7 Emergency Stop

Press any red emergency stop button to stop the machine immediately. All motion stops (Category 1), drive torque is removed and the pneumatic system is exhausted. The hot-melt unit stays powered so the glue does not solidify, and the vertical brake holds the pick-and-place head.

To restart: remove the cause of the emergency, pull out the emergency stop button, press **Reset** on the operator panel, then press **Start**. Releasing the button alone never restarts the machine.

⚠ WARNING

Automatic restart after a power interruption or fault reset.

Machine parts can start moving without warning and cause serious injury.

- Do not rely on a stopped machine being safe; lock out before entering.
- Confirm that auto-restart parameters are disabled unless the risk assessment allows them.

1.8 Safety Signs and Labels

Label	ISO 7010	Location	Meaning
Electrical hazard	W012	Electrical cabinet door, drive covers	Hazardous voltage inside; qualified persons only
Hot surface	W017	Glue tank, hoses and glue heads	Burn hazard; let cool before touching
Crushing of hands	W024	Case erector and flap closer	Keep hands clear of moving parts
Entanglement	W026	Infeed and discharge belt ends	Keep guards closed; no loose clothing
Overhead load	W015	Pick-and-place gantry	Never stand under the raised head
Read the manual	M002	Operator panel	Read the manual before operation
Lifting point	—	Four lifting eyes per section	Attach slings here only
Disconnect before servicing	M021	Main disconnect Q0	Lock out before maintenance

Replace any label that is damaged or unreadable. The complete label set is part no. SA-9001 (Section 10.1).

1.9 Personal Protective Equipment

Task	Eyes	Hands	Feet	Other
Production	Safety glasses	—	Safety shoes	Hearing protection if above 80 dB(A)
Glue refill / glue head cleaning	Safety glasses	Heat-resistant gloves	Safety shoes	Long sleeves
Format changeover	Safety glasses	Cut-resistant gloves	Safety shoes	—
Mechanical maintenance	Safety glasses	Mechanic's gloves	Safety shoes	Bump cap inside the guard
Electrical work	Face shield	Voltage-rated gloves	Safety shoes	Arc-rated clothing as labeled

1.10 Energy Isolation and Lockout/Tagout

Lock out the machine before any cleaning, maintenance or jam clearing inside the guards. Opening a guard door is **not** a lockout. The SAMPLE-A has the energy sources and isolation points listed below; their locations are shown in Figure 1-1.

Isolation points

ID	Device	Location	Energy isolated	Lock	How to verify zero energy
Q0	Main disconnect, rotary, lockable	Electrical cabinet door	480 V supply to the whole machine except the hot-melt unit	Personal padlock in handle	Try **Start**; measure 0 V at drive input terminals
Q5	Hot-melt unit disconnect	Hot-melt unit side panel	480 V to hot-melt heaters and pump	Personal padlock	Unit display dark; heaters cooling
V0	Lockable ball valve with exhaust	Air inlet module, infeed end	6 bar compressed air	Valve lockout cover	Gauge on filter regulator reads 0 bar
—	Stored energy: drive capacitors	U1–U3, servo drive A1	DC bus voltage	—	Wait 5 min; DC-bus LED off; measure
—	Stored energy: raised head	Pick-and-place head	Gravity	Block with support bar SA-0090	Head resting on the support bar

ID	Device	Location	Energy isolated	Lock	How to verify zero energy
—	Stored energy: hot glue	Tank, hoses, guns	Heat, molten adhesive under pressure	—	Pressure relieved; surface below 50 °C

Lockout procedure

1. Tell everyone affected that the machine will be locked out.
   → Operators and maintenance staff know the machine is going out of service.
2. Press **Stop** and wait for the end of the cycle.
   → Stack light red; all stations at home.
3. Lower the pick-and-place head to its rest position from the *Manual Control* screen.
   → Head rests on its lower end stop.
4. Turn main disconnect Q0 to OFF.
   → Operator panel goes dark.
5. Close air valve V0.
   → Dump valve exhausts; air gauge falls to 0 bar.
6. If you will work on the glue system, turn hot-melt disconnect Q5 to OFF.
   → Hot-melt display goes dark.
7. Attach your personal lock and tag to each isolation point you used.
   → Each point carries one lock per person working.

⚠ DANGER

Drive and power-supply capacitors stay charged after power is removed.

Touching DC bus terminals before discharge will cause death or severe injury.

- After isolation wait at least the stated discharge time (5 min).
- Measure the DC bus before work and confirm it is below 50 V.

8. Wait 5 minutes for the drive capacitors to discharge.
   → DC-bus LEDs on all drives are off.
9. Verify zero energy: press **Start**, read the air gauge, and measure voltage at the drive input terminals.
   → Nothing moves; gauge 0 bar; 0 V measured.
10. Do the work.

Returning to service

1. Remove tools and materials from inside the guard.
   → Guarded area is clear.
2. Refit all fixed guards and close all doors.
   → All doors closed; fixed guards bolted.
3. Make sure nobody is inside the guarded area.
   → Visual check done from all sides.
4. Remove your lock and tag. Only the person who fitted a lock may remove it.
   → All isolation points free.
5. Open V0, then switch on Q0 (and Q5 if used).
   → Operator panel starts; alarm A103 asks for reset.
6. Test the emergency stop at the operator panel before production (Section 9.4).
   → Machine cannot start while E2 is pressed.

Group lockout: when more than one person works on the machine, use a group lock box. Each person places a personal lock on the box. The isolation points are locked with the group locks, whose keys are kept in the box.

1.11 Residual Risks

The following risks remain despite the guards and safety functions. They are listed by zone; observe the instructions in each message.

Zone	Residual risk	When it can occur	Protective measure
Z1	Drawing-in at infeed rollers	Clearing fallen products at the tunnel	Stop the machine; use the hook tool; never reach in while running
Z2	Falling head or products	Brake failure during maintenance	Rest the head on the support bar before working below it
Z2	Crushing by gantry in Setup mode	Jogging with door D1 open	Reduced speed 10 mm/s; enabling device; stay outside the travel path
Z3	Crushing by swing arm	Clearing a misfed blank	Lock out; light curtain does not protect inside the guard
Z4	Burns from hot-melt	Refilling, nozzle cleaning	Heat-resistant gloves, safety glasses; tank lid only from the side
Z4	Residual air pressure	Cylinders stopped mid-stroke	Check the gauge reads 0 bar after lockout
Z6	Electric shock	Work in the cabinet	Qualified persons; lockout; test before touch
All	Lifting change parts	Changeover	Two-person lift above 23 kg

⚠ DANGER

Hazardous voltage inside electrical enclosures and terminal boxes.

Contact with live parts will cause death or severe burns.

- Only qualified electrical workers may open the enclosure.
- Switch off and lock the main disconnect before opening.
- Verify absence of voltage with a rated tester on every conductor before touching.

⚠ WARNING

Crushing points at the case erector, pick-and-place head and flap closer.

Body parts caught in the gap can be crushed or amputated.

- Keep hands and body clear of the motion path while the machine is energised.
- Use setup mode with reduced speed and an enabling device for close-in work.
- Block or lower gravity-loaded parts before reaching in.

⚠ WARNING

Rotating rollers and belts at the infeed and discharge conveyor ends.

Hair, clothing, gloves or jewellery can be drawn in, causing severe injury or death.

- Keep guards closed while parts rotate.
- Tie back long hair; no loose clothing, jewellery or gloves near rotating parts.
- Wait for complete standstill before opening guards.

⚠ WARNING

The pick-and-place head is held up by a motor brake and can drop if the brake is released or fails.

A falling load can crush or kill.

- Mechanically block or lower vertical axes before maintenance.
- Never stand under suspended loads.
- Test holding brakes at the specified interval.

⚠ WARNING

Hot-melt tank, hoses and glue heads operate at up to 180 °C.

Touching hot parts or molten glue causes severe burns.

- Wear heat-resistant gloves and safety glasses when refilling glue or cleaning nozzles.
- Let the glue unit cool for 30 minutes before maintenance.

⚠ WARNING

Compressed air (6 bar) in the vacuum, erector and flap-folding cylinders.

Sudden movement of cylinders, whipping hoses or flying parts can cause injury.

- Close and lock the air isolation valve and exhaust residual air before work.
- Secure loads held by cylinders.
- Never point compressed air at people.

⚠ CAUTION

Change parts and bundles of case blanks can weigh more than 23 kg.

Can cause back and muscle injuries.

- Use lifting aids for loads above 23 kg (or local limit).
- Lift with two people where aids are not available.

1.12 Noise, Vibration and Emissions

The A-weighted emission sound pressure level at the operator position is 74 dB(A) (uncertainty K = 3 dB), measured to ISO 11204 at 20 cases/min with 300 × 250 × 200 mm cases. The peak C-weighted value does not exceed 130 dB. Hand-arm vibration is not relevant; whole-body vibration at the operator position is below 0.5 m/s². Hot-melt adhesive can release small amounts of fumes at application temperature; provide general room ventilation of at least 4 air changes per hour.

1.13 Fire Prevention

Hot-melt adhesive and corrugated board are combustible. Keep the area around the glue unit free of carton scraps, never heat the adhesive above 200 °C, and keep a CO_2 or dry-powder extinguisher within 10 m of the machine. Never use water on molten glue.

⚠ WARNING

Combustible adhesive and carton dust near heated glue components.

Fire or explosion can cause death, serious injury and property loss.

- Use only non-flammable coolant unless a fire suppression system is fitted.
- Do not leave the machine unattended while machining flammable materials.
- Keep a suitable extinguisher nearby.

1.14 Cybersecurity

The controller and operator panel connect to the plant network through a managed switch with a firewall. Remote access (option R) is disabled by default and can only be opened from the operator panel at Engineer level for a limited time. Operator panel functions are protected by four user levels (Section 6.3). Change the default passwords during commissioning, and do not connect USB devices other than those used for recipe backup.

2 Machine Description and Theory of Operation

2.1 What the Machine Does

The SAMPLE-A takes a steady stream of sealed products and turns it into closed, glued shipping cases ready for the pallet. Flat corrugated blanks go in at one side, products come in from the upstream line, and a finished, square, sealed case leaves at the other end, up to 20 times a minute.

Every case passes through five stations in a fixed order: the **infeed conveyor** spaces and counts the products, the **collation and pick-and-place** station groups them into the case pattern and lowers them into an open case, the **case erector** forms each case from a flat blank, the **glue and flap closing** station seals it, and the **discharge conveyor** hands it on to the downstream line. Two support systems serve all stations: the **air and vacuum system** and the **electrical and control system**. Sections 2.4 to 2.10 describe how each works; Section 2.11 describes how they work together in one cycle.

2.2 Machine Layout and Product Flow

Products flow from left to right as seen from the operator panel. Stations 2 to 4 are inside the interlocked guard; the infeed and discharge conveyors pass through tunnel openings in the guard.

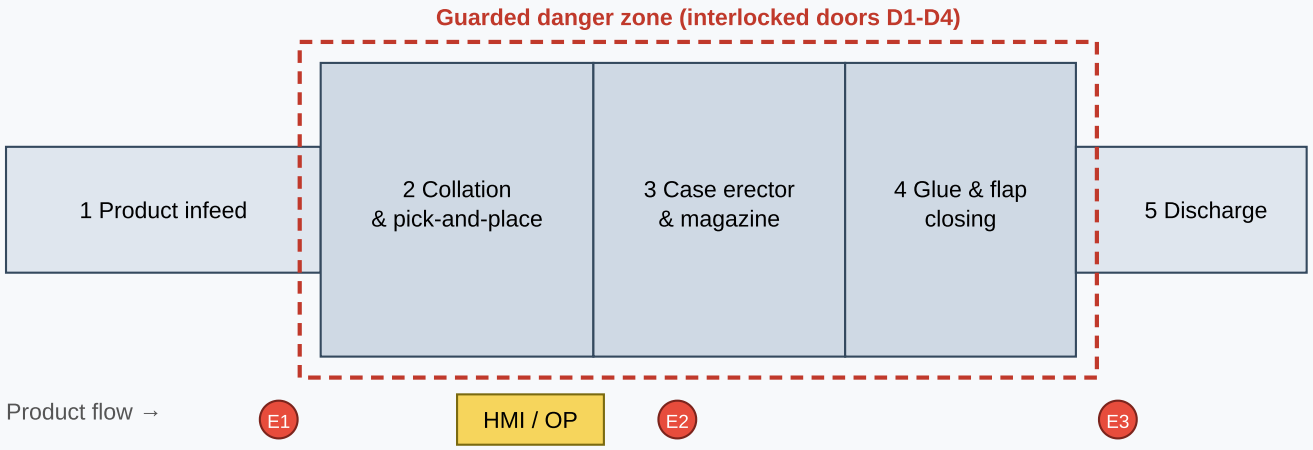


Figure 2-1. Machine layout in product-flow order. Numbers match the station sections that follow.

2.3 Main Assemblies and Their Functions

No.	Assembly	What it does	Section
1	Infeed conveyor	Brings products in single file, spaces and counts them	2.4
2	Collation and pick-and-place	Groups products into the case pattern and loads them into the open case	2.5
3	Case erector and blank magazine	Stores flat blanks and forms one open case per cycle	2.6
4	Glue and flap closing	Applies hot-melt adhesive, folds and presses the flaps shut	2.7
5	Case discharge conveyor	Moves the sealed case out and hands it to the downstream line	2.8
6	Air and vacuum system	Supplies clean air to cylinders and vacuum to the grippers	2.9
7	Electrical cabinet and controls	Controller, drives, safety controller, power distribution	2.10
8	Operator panel	Touch screen, Start / Stop / Reset buttons, emergency stop E2	Ch. 6

2.4 How the Infeed Conveyor Works

The infeed conveyor is the machine's entry point. It takes products from the upstream line, opens up a regular gap between them so they can be counted reliably, and holds products back whenever the collation station cannot accept more.

Main components	Flat-top belt conveyor (2,350 mm) driven by gearmotor M10 on variable-speed drive U1; adjustable side guides; metering belt at the discharge end driven from the main belt through a fixed pulley ratio
How it moves	Main belt runs continuously at line speed; the metering belt runs about 15 % faster to open a gap of 20–40 mm between products
Sensors	Product sensor B10 at the metering belt counts each product; backup sensor B11, 1 m upstream, detects a queue forming
Interlocks / permissives	Runs only in Execute with air pressure OK; pauses when the collation lane is full; stops on any red alarm
Operator interaction	Set side-guide width at changeover (color-coded scale); clear fallen products at the tunnel opening after stopping
Program reference	Infeed logic, rungs 0–13

When the machine starts, the main belt runs first; it carries the metering belt with it through the pulley ratio, so the gap between products grows as each one crosses onto the faster belt. Each product breaking the product-sensor beam adds one to the count for the current pattern. **Inferred – please confirm** *Belt speed ratio inferred from CAD pulley sizes.*

When things go wrong. If the backup sensor stays blocked for 3 seconds the infeed slows to half speed and amber alarm A111 appears; it returns to full speed by itself when the queue clears. If the drive trips, for example on a jammed product, the infeed stops with red alarm A110. If the count disagrees with the number of products the collation actually received, the machine finishes the case and stops with A112 so a wrong product count never reaches the customer.

Adjustment points. Side guides (handwheel with scale), sensor brackets (slotted, see Section 9.8), belt tension (take-up screws at the idler end, Section 9.5).

⚠ WARNING

Rotating rollers and belts at the infeed conveyor ends and the metering transfer.

Hair, clothing, gloves or jewellery can be drawn in, causing severe injury or death.

- Keep guards closed while parts rotate.
- Tie back long hair; no loose clothing, jewellery or gloves near rotating parts.
- Wait for complete standstill before opening guards.

2.5 How the Collation and Pick-and-Place Works

This station builds the product pattern for one case (for example 3 rows × 4 products) and places the complete layer into the open case in one move. It is the heart of the machine and sets the overall cycle rate.

Main components	Lane divider and servo-driven collation fingers; two-axis servo gantry (horizontal and vertical) with a vacuum pick head; vacuum generator VG1 with vacuum switch B31
How it moves	Collation fingers index one row at a time; the gantry travels up to 600 mm horizontally and 350 mm vertically along the path in Figure 2-2
Sensors	Row-complete sensor B12; vacuum switch B31 confirms grip; servo position feedback on all axes
Interlocks / permissives	Pick only when a full pattern is collated; place only when the case erector reports a case open and square; head returns to the safe height before any horizontal travel
Operator interaction	Change collation fingers and pick-head plate at changeover; jog axes in Setup mode from the *Manual Control* screen
Program reference	Pick-and-place logic, rungs 0–27

Products are pushed into lanes and stopped by the collation fingers until each row is complete. When the row-complete sensor sees the last product of a row, the fingers index one row pitch and the next row starts. When the last row arrives, the pick head descends (1), switches on vacuum and waits for the vacuum switch to confirm a grip before lifting (2). It then lifts to the safe travel height and travels to the case (3), lowers the pattern to 10 mm above the case bottom (4) and releases it, then lifts and returns (5) while the next pattern is already being collated.

When things go wrong. If the grip is lost during travel, the head stops, sets the products back down at the nearest safe position and raises alarm A130. If either axis falls more than 5 mm behind its commanded position (for example because the head has hit an obstruction) the gantry stops with A120 or A121. The vertical axis has a holding brake, so the head stays in place on power loss; the brake is tested automatically every time the axes are homed (A123 if it fails).

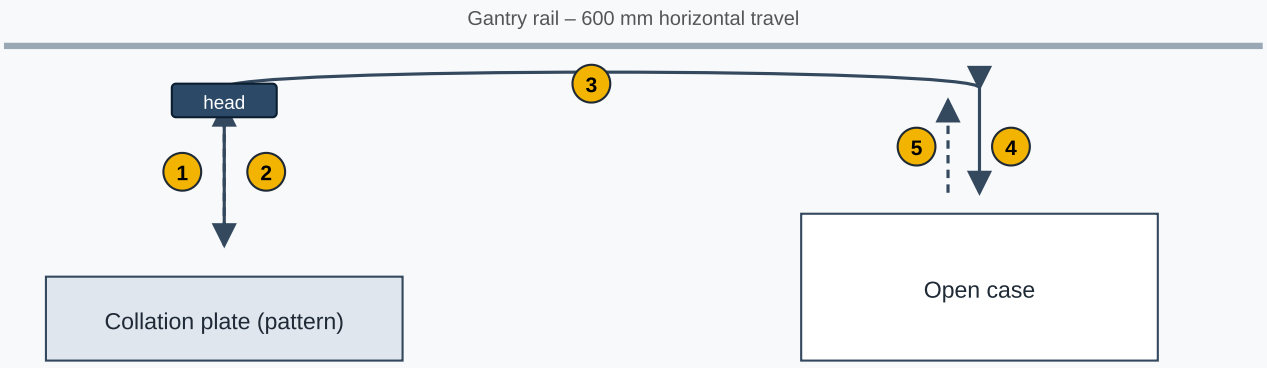


Figure 2-2. Pick-and-place motion: (1) lower and grip, (2) lift, (3) travel at safe height, (4) lower into case and release, (5) lift and return.

⚠ WARNING

The pick-and-place head is held up by a motor brake and can drop if the brake is released or fails.

A falling load can crush or kill.

- Mechanically block or lower vertical axes before maintenance.
- Never stand under suspended loads.
- Test holding brakes at the specified interval.

2.6 How the Case Erector Works

The case erector forms a square, open-topped case from a flat corrugated blank, ready for loading. One case is erected for every product pattern.

Main components	Gravity-fed blank magazine (up to 150 blanks); servo-driven swing arm with four vacuum cups; forming plow; bottom-flap folders driven by pneumatic cylinders
How it moves	Swing arm pulls a blank from the magazine and rotates it 90°; the forming plow opens the blank into a square tube; folders close the minor and major bottom flaps (Figure 2-3)
Sensors	Blank-present sensor B20 and magazine-low sensor B21; vacuum switch B22 on the swing arm; case-square sensor B23; case-in-position sensor B25; light curtain LC1 at the magazine
Interlocks / permissives	Starts a new case only when the previous case has left the loading position; holds while the light curtain is interrupted
Operator interaction	Reload blanks from outside the guard while running; adjust magazine guides at changeover
Program reference	Case erector logic, rungs 0–21

The swing arm presses its vacuum cups against the front blank, waits for the vacuum switch, then rotates. As it swings, the blank meets the forming plow, which opens it into a square tube. Two cylinders fold the minor flaps and a third pushes the major flaps into position, so the case bottom is closed and glued in the glue station. The case-square sensor checks the diagonal before the case is reported open. **Inferred – please confirm** *Bottom flaps glued at station 4, inferred from CAD glue-gun position.*

When things go wrong. If no vacuum is reached within 1.5 seconds, the arm retries once and then raises alarm A135. A case that is not square (A136) is held so it can be removed before it is loaded. A cylinder that does not reach its end position within 0.8 s raises A137, which usually means low air pressure or a sticking valve.

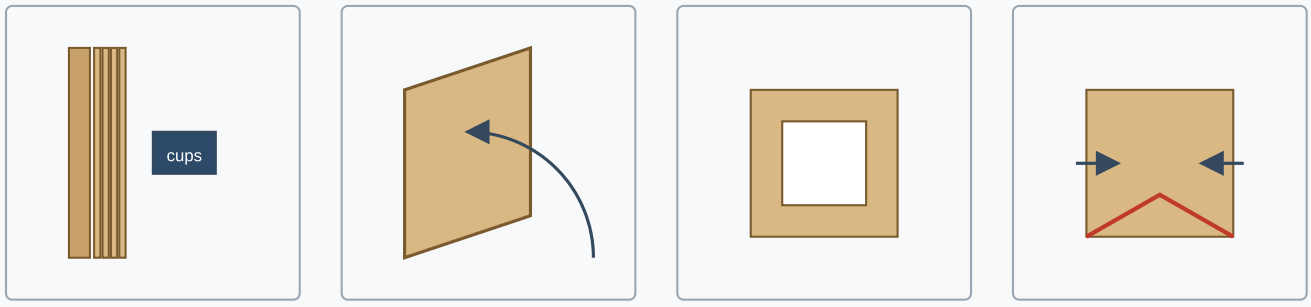


Figure 2-3. Case forming in four steps.

⚠ WARNING

Crushing points between the swing arm, forming plow and flap folders of the case erector.

Body parts caught in the gap can be crushed or amputated.

- Keep hands and body clear of the motion path while the machine is energised.
- Use setup mode with reduced speed and an enabling device for close-in work.
- Block or lower gravity-loaded parts before reaching in.

2.7 How the Glue and Flap Closing Works

This station seals the case. It applies hot-melt adhesive to the flaps, folds the top and bottom flaps closed and holds the case under pressure until the glue sets.

Main components	Hot-melt unit with 10 kg tank, heated hoses and four glue guns (two top, two bottom); top-flap plows and tucker cylinder; side compression belts on drive U2
How it moves	Case moves through on the compression belts; fixed plows fold the leading, trailing and side flaps; glue guns fire as each flap passes
Sensors	Case-entry sensor B40 triggers the glue pattern; encoder on the compression belt times each glue dot; glue-ready, level and fault signals from the hot-melt unit
Interlocks / permissives	Cases are not released into this station unless the glue is at temperature; glue guns fire only while a case is moving past them
Operator interaction	Refill glue (amber alarm A141); check glue dots on the first cases after a restart
Program reference	Glue and flap logic, rungs 0–19

When the case-entry sensor sees the leading edge, the control counts encoder pulses and fires each glue gun at the programmed distance, so the glue pattern stays in the same place on the flap at any speed (Figure 2-4). The opening time of each gun valve is compensated, so dots do not drift when the speed changes. The plows fold the flaps onto the glue, and the compression belts hold the case square for the recipe time (normally 1.5 seconds) while the adhesive sets.

When things go wrong. If the glue temperature drops more than 10 °C below set point (A140), the station finishes the case in progress and then holds new cases until the glue recovers. A low tank level (A141) is a warning only; about 30 minutes of production remain. A fault reported by the hot-melt unit (A142) stops the machine at the end of the cycle.

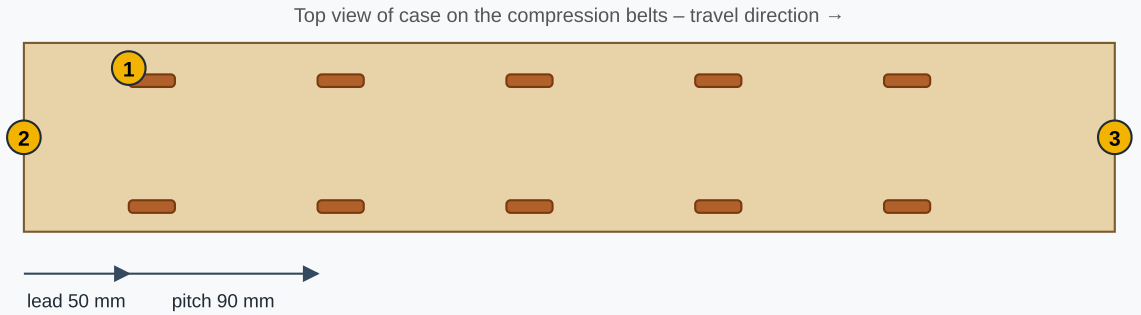


Figure 2-4. Glue pattern on one flap: (1) first dot 50 mm from the leading edge, (2) leading edge seen by the case-entry sensor, (3) trailing edge. Five dots per line at 90 mm pitch (recipe values).

⚠ WARNING

Hot-melt tank, hoses and glue guns operate at up to 180 °C.

Touching hot parts or molten glue causes severe burns.

- Wear heat-resistant gloves and safety glasses when refilling glue or cleaning nozzles.
- Let the glue unit cool for 30 minutes before maintenance.

2.8 How the Case Discharge Works

The discharge conveyor carries the sealed case out of the guard and hands it to the downstream line, and it tells the SAMPLE-A when the downstream line is backing up.

Main components	Powered roller conveyor (1,200 mm) with gearmotor M50 on drive U3
How it moves	Runs continuously while cases are present; stops 30 seconds after the last case leaves
Sensors	Case-exit sensor B50; downstream-full signal from the next machine
Interlocks / permissives	When the downstream line is full, the conveyor stops and the machine finishes the case in progress, then pauses (Suspended)
Operator interaction	None in normal production; clear the downstream conveyor if alarm A160 appears
Program reference	Discharge logic, rungs 0–9

The machine resumes by itself when the downstream-full signal clears, so the operator does not need to press **Start** after a short downstream stop. A case that does not clear the exit sensor within 4 seconds raises A161 (discharge jam), usually caused by an open flap catching on the guard.

2.9 How the Air and Vacuum System Works

The air system supplies clean, dry compressed air to every cylinder and to the two vacuum generators, and it makes the pneumatic system safe when the machine stops.

Main components	Lockable inlet valve V0; filter regulator with 5 µm filter and gauge; soft-start and dump valve Y01; pressure switch B30; valve manifold with six valves; vacuum generators VG1 (pick head) and VG2 (swing arm)
How it moves	Air flows from V0 through the filter regulator and soft-start valve to the manifold; each valve drives one actuator or vacuum generator (see Appendix 12.3)
Sensors	Pressure switch B30 (5 bar); vacuum switches B22 and B31; end-position sensors on every cylinder
Interlocks / permissives	No start below 5 bar; Y01 drops on every E-stop or door opening and exhausts the system
Operator interaction	Check the gauge reads 6 bar at start of shift; drain condensate weekly
Program reference	Air and vacuum logic, rungs 0–7

After a safety reset the soft-start valve lets pressure build up slowly over about 2 seconds, so cylinders move gently to their start positions. Vacuum is made by venturi generators that need only compressed air; each has a vacuum switch so the control knows whether a grip is good before anything is lifted.

2.10 How the Electrical and Control System Works

The control system coordinates all stations, runs the safety functions and lets the operator control the machine.

Main components	Main disconnect Q0; drives U1 (infeed), U2 (compression belts), U3 (discharge); four-axis servo drive A1; 24 V DC power supply; controller A10 with integrated safety controller; operator panel A20; line interface X10/X11
How it moves	Servo axes follow motion profiles calculated for each recipe; conveyor drives follow speed set points
Sensors	All machine sensors connect to remote I/O modules over the industrial Ethernet fieldbus
Interlocks / permissives	Safety controller removes drive torque (STO) and drops the dump valve; standard control cannot override it
Operator interaction	Operator panel only; cabinet access for electrical maintenance
Program reference	All logic; see Appendix 12.7

Standard control and safety control run in the same controller but in separate, protected tasks. The safety task is certified and locked with a signature; any change to it requires the manufacturer. The electrical overview is in Appendix 12.2.

2.11 Sequence of Operation

One automatic cycle, from the product's point of view. Stations work in parallel: while one case is being loaded, the next blank is already being erected and the previous case is being sealed. At 20 cases/min each cycle takes 3.0 seconds (Figure 2-6).

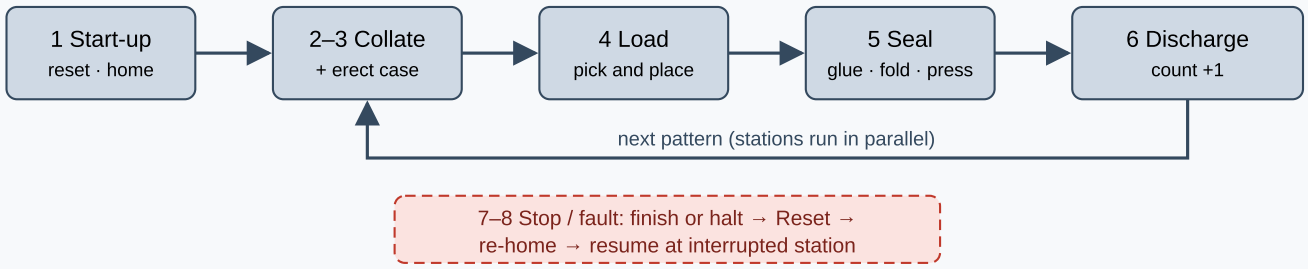


Figure 2-5. Automatic cycle. The dashed path shows what happens after a stop or fault.

1. **Start-up.** After **Reset** and **Start**, the gantry moves to its home position, the case erector makes the first case, and the infeed belts start.
2. **Collate.** The infeed spaces and counts products; the collation fingers build rows until one full pattern is ready.
3. **Erect.** In parallel, the swing arm pulls a blank, forms it into a case and folds the bottom flaps. The case waits in the loading position.
4. **Load.** With a full pattern and an open case both ready, the pick head grips the pattern by vacuum, travels and lowers it into the case.
5. **Seal.** The loaded case is pushed onto the compression belts; glue is applied and the flaps are folded and pressed shut.
6. **Discharge.** The sealed case leaves on the discharge conveyor and the case count increases by one.
7. **Cycle stop.** Pressing **Stop** lets the current pattern finish loading and the last case clear the sealer, then all stations stop in their home positions.
8. **After a fault or E-stop.** Motion stops immediately. After the cause is removed and **Reset** is pressed, the machine re-homes the gantry and resumes from the station that was interrupted.

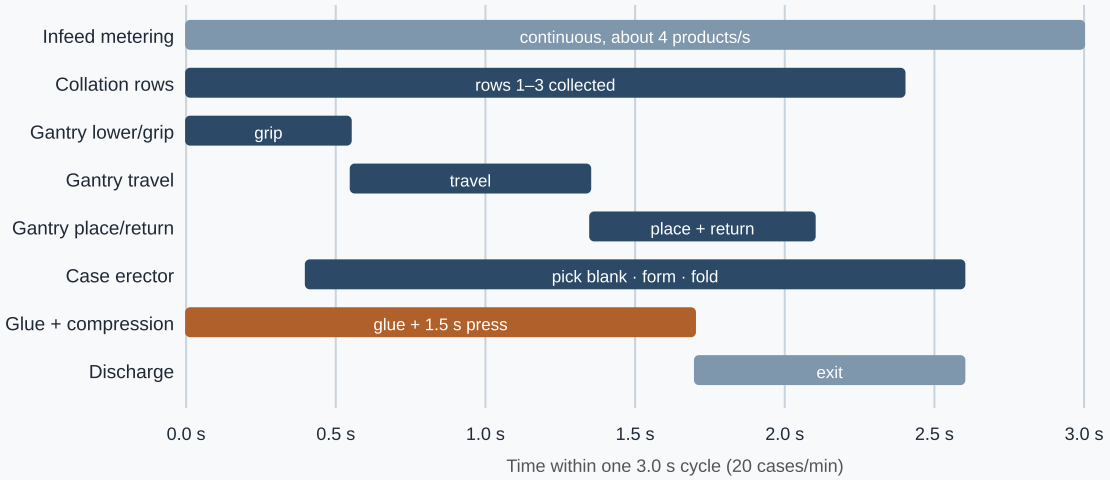


Figure 2-6. Timing of one cycle at 20 cases/min (recipe F02). Bars show when each station is active.

2.12 Interlocks, Permissives and Machine States

The machine follows a state model based on ISA-88 / ISA-TR88.00.02. The current state is shown on the *Home* screen and determines which buttons are accepted.

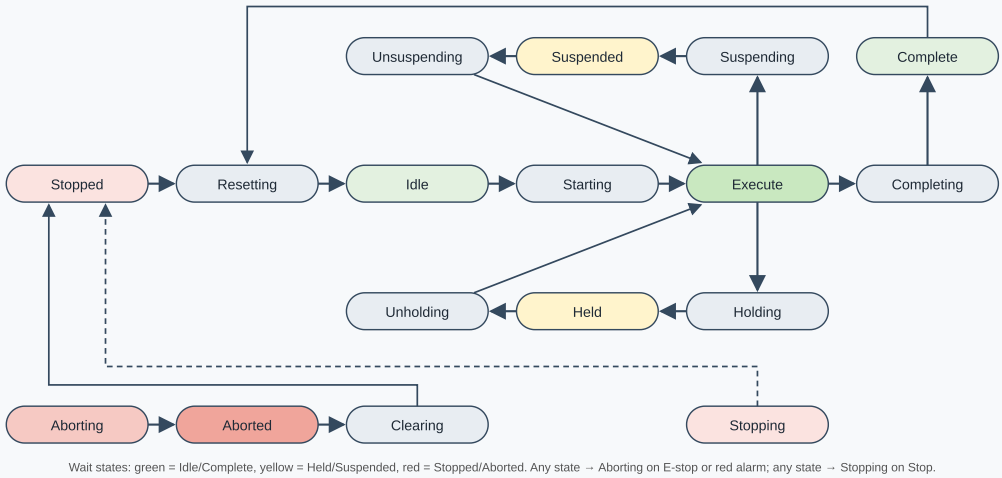


Figure 2-7. Machine state model. Acting states (white) end by themselves; wait states (shaded) need a command or a condition.

State	What the machine is doing	Leaves the state when
Stopped	Powered, not moving, safety circuit may need reset	**Reset** pressed
Resetting	Re-homing axes, building air pressure	Homing complete → Idle
Idle	Ready to run; all permissives met	**Start** pressed
Starting	Ramping conveyors, making first case	Stations ready → Execute
Execute	Producing cases	**Stop**, fault, or downstream full
Suspending / Suspended	Downstream full or upstream starved; finishes current case and waits	Condition clears → Unsuspending → Execute
Holding / Held	Operator pressed **Hold** or amber condition needs attention (glue cold)	**Start** or condition clears → Unholding
Completing / Complete	Batch count reached; empties the machine	**Reset**
Stopping	Cycle stop in progress	All stations home → Stopped
Aborting / Aborted	Emergency stop or red alarm; immediate stop	Cause removed, **Reset** → Clearing → Stopped

Interlocks and permissives

Condition	What it does	Stations affected	How to clear
Guard door open or E-stop pressed	Stops all motion; machine cannot start	All	Close doors / release E-stop, press **Reset**
Air pressure below 5 bar	Prevents start; stops after current cycle	Erector, pick-and-place, flaps	Restore plant air
Glue not at temperature	Holds new cases before the sealer	Glue and flap closing	Wait for heat-up
No open case in loading position	Pick head waits with the pattern	Pick-and-place	Automatic; check erector if it persists
Collation full	Infeed pauses	Infeed	Automatic

Condition	What it does	Stations affected	How to clear
Downstream line full	Discharge stops; machine suspends after current case	Discharge, then all	Automatic when downstream clears
Axes not referenced	Prevents Automatic	Pick-and-place, erector	**Reset**, then **Home**
Recipe invalid or not loaded	Prevents start	All	Load a valid recipe
Magazine below 20 blanks	Amber warning only	Case erector	Reload blanks

2.13 Control Logic Description

This section explains what the control program makes the machine do, grouped by machine function. Each group summarizes several related rungs of the program; the small reference shows which rungs. All 200 rungs of the program are accounted for, as listed in Appendix 12.7.

Modes, machine states and starting

Function	What the logic does	Rungs
Mode selection	Reads the mode key switch. Automatic is allowed only with all guard doors closed; Setup limits every axis to jog speed and enables the hold-to-run function; Clean allows only slow conveyor movement.	0–3
Machine state model	Moves the machine through the standard packaging-machine states in Section 2.12 (Stopped, Resetting, Idle, Starting, Execute, Holding, Suspended, Stopping, Aborting and the rest) and makes sure only one state is active at a time.	4–11
Start permissives	Combines safety reset, air pressure, glue ready, all axes referenced and a valid recipe into a single Ready condition. **Start** is accepted only when Ready is true.	12–15
Homing after reset	Raises the gantry first, then moves it across to its home position, then homes the collation fingers and the swing arm. No other station may start before homing is complete.	16–19
Stack light and counters	Drives the stack-light colors from the machine state and counts good cases, rejected cases and shift totals.	20–23

Safety status monitoring

Function	What the logic does	Rungs
Safety device status	Copies the status of each emergency stop, guard door and the light curtain from the safety controller to the operator panel so the screen can show exactly which device stopped the machine. The safety functions themselves run in the certified safety controller (Section 1.6).	0–4
Safe Torque Off feedback	Compares the drive Safe Torque Off feedback with the safety relay state; a mismatch for more than 200 ms raises alarm A104 and blocks reset.	5–8
Reset request	Lights the blue **Reset** button when the safety circuit can be reset and passes the reset command to the safety controller only on release of the button.	9–11

Infeed conveyor

Function	What the logic does	Rungs
Running the infeed	Runs the main belt while the machine is in Execute; the metering belt follows through the fixed pulley ratio. Pauses the infeed when the collation lane is full.	0–5
Counting products	Counts each product passing the metering sensor toward the current pattern and resets the count when the pattern is picked. A count that disagrees with the collation rows raises A112.	6–9
Backup protection	Slows the infeed to half speed after 3 s of queue at the backup sensor (A111) and stops it with alarm A110 if the drive reports a fault.	10–13

Collation and pick-and-place

Function	What the logic does	Rungs
Building the pattern	Indexes the collation fingers one row at a time and signals Pattern Ready when the last row of the recipe is complete. Raises A125 if a pattern is not complete within 10 s while products are arriving.	0–7
Pick, travel and place	Runs the pick-place-return motion as a step sequence: lower, vacuum on, confirm grip, lift to safe height, travel, lower into case, release, lift, return. Each step waits for the axis to arrive before the next begins.	8–21
Grip-loss recovery	If vacuum is lost during travel, stops the gantry, sets the pattern back down at the nearest safe position and raises alarm A130.	22–24
Brake and following-error monitoring	Holds the vertical brake applied whenever the axis is not enabled and tests the brake at every homing; raises A120 or A121 if an axis lags its commanded position and A123 if the brake test fails.	25–27

Case erector

Function	What the logic does	Rungs
Picking a blank	Starts a new case once the loading position is empty; waits up to 1.5 s for vacuum at the swing-arm cups and retries once before raising A135.	0–8
Forming and bottom-flap folding	Times the minor- and major-flap cylinders to the swing-arm position and confirms the case is square before reporting Case Open; raises A136 or A137 when the case is not square or a cylinder does not reach its end position.	9–17
Magazine and light curtain	Raises the magazine-low warning at 20 blanks (A170), requests operator action when empty (A171) and holds the erector while the light curtain is interrupted (A180).	18–21

Glue and flap closing

Function	What the logic does	Rungs
Glue readiness	Holds new cases before the station until the hot-melt unit reports temperature; reports A140, A141 and A142 from the hot-melt unit's status signals.	0–3
Glue pattern timing	Starts a glue pattern at the case leading edge and fires each gun at its programmed distance using the compression-belt encoder, so dots stay in place at any speed. Compensates for valve opening time.	4–15
Compression and release	Keeps the compression belts running until the case has been under pressure for the recipe time (normally 1.5 s), then releases it to discharge.	16–19

Case discharge

Function	What the logic does	Rungs
Discharge conveyor control	Runs while cases are present and for 30 s after; raises A161 if a case takes more than 4 s to clear the exit sensor.	0–5
Downstream handshake	Stops and moves the machine to Suspended when the downstream line signals full (A160), then resumes automatically when the signal clears.	6–9

Air supply and vacuum

Function	What the logic does	Rungs
Air pressure monitoring	Raises A150 when supply pressure falls below 5 bar for more than 1 s; the machine finishes the current cycle and stops.	0–3
Soft-start and dump valve	Opens the soft-start valve after a safety reset so pressure builds up gradually, and drops it at every emergency stop or door opening so all cylinders exhaust.	4–7

Recipes and changeover

Function	What the logic does	Rungs
Recipe load and validation	Loads the selected format, checks every value against its limits and refuses to load a recipe with a value outside its range.	0–7
Changeover checklist	Shows the changeover steps for the new format and holds the machine in Stopped until each step is confirmed on the operator panel.	8–12
Speed scaling	Converts the cases-per-minute set point into belt speeds and gantry motion times for the selected format.	13–15

Alarms and diagnostics

Function	What the logic does	Rungs
Alarm detection	Turns each fault condition into the alarm codes listed in Section 8.1, with a time stamp and the station concerned.	0–27
Alarm classes and reset	Latches red alarms until **Reset**, lets amber alarms clear by themselves when the cause is gone, and shows blue alarms as operator requests.	28–31

Maintenance counters

Function	What the logic does	Rungs
Hour counters and reminders	Counts running hours and cycles for each station and shows a blue maintenance reminder (A190) when a task in Section 9.2 is due.	0–7

2.14 Identification and Nameplate

The nameplate is on the electrical cabinet door. It shows: manufacturer, model, serial number, year of manufacture, supply voltage and frequency, full-load current, short-circuit current rating, air pressure, machine mass, and compliance marks. Quote the model and serial number when contacting service.

2.15 Options and Accessories

Option	Description
T – Tape sealer	Top-and-bottom tape head instead of hot-melt; the sealing step is otherwise the same
L – Case printer interface	Prints or labels each case on the discharge conveyor
R – Remote diagnostics gateway	Secure, time-limited remote access for the manufacturer's service team
B – Batch counter	Stops the machine in Complete after a set number of cases

3 Transport, Handling and Storage

3.1 Transport Dimensions and Masses

The machine ships in two sections on skids. The operator panel and stack light are packed separately in a crate.

Section	Mass	Dimensions (L × W × H)	Lifting
A – Infeed and collation	1,150 kg	2,400 × 1,500 × 2,100 mm	Forklift from side, or 4 lifting eyes
B – Erector, glue and discharge	1,450 kg	2,900 × 1,600 × 2,100 mm	Forklift from side, or 4 lifting eyes
Crate – operator panel, spares	85 kg	1,200 × 800 × 900 mm	Pallet truck

3.2 Lifting

Lift only with a forklift of at least 3,000 kg capacity from the marked side, or with a crane and four slings of 1,000 kg working load each attached to the lifting eyes. The center of gravity is marked on each section.

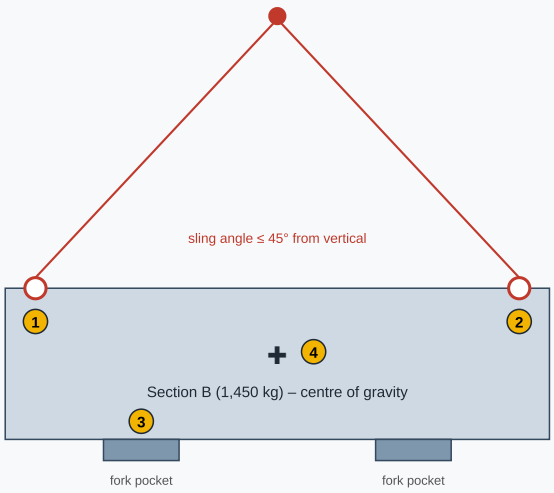


Figure 3-1. Lifting points on section B: (1, 2) lifting eyes, (3) fork pockets, (4) center-of-gravity mark. Section A is the same.

1. Check that the transport locks (red brackets) are fitted at the gantry, swing arm and folder.
   → Six red brackets visible.
2. Clear the transport route and mark off the area under the load.
   → Nobody in the area.

⚠ WARNING

Heavy machine or modules during lifting and transport, including off-centre centre of gravity.

Tip-over or a dropped load can cause death or serious injury and damage the machine.

- Use only the marked lifting points and lifting gear rated for the stated mass.
- Fit transport locks before moving.
- Keep people out of the lift area.

3. Insert the forks fully into the fork pockets, or attach slings to all four lifting eyes.
   → Forks extend beyond the far side; sling angle no more than 45° from vertical.
4. Lift 100 mm and check the load is level.
   → Section hangs level and stable.
5. Move slowly to the installation position and lower onto the floor.
   → Section stands on its feet; forks or slings slack.

3.3 Unpacking and Inspection on Receipt

Check	Requirement / expected result	OK
Packing list	All items on the packing list present	☐
Tilt and shock indicators	Not triggered; if triggered, note on the delivery receipt	☐
Guards and doors	No dents; glass panels unbroken	☐
Cabinet	Door closes; no loose components inside	☐
Conveyor belts	No cuts or creases	☐
Spare parts crate	Sealed; contents match the list	☐

Report damage to the carrier and to the manufacturer within 48 hours. Remove the red transport locks only after the machine is in its final position (Section 4.2).

3.4 Storage

Period	Requirements
Up to 6 months	Indoors, 0–45 °C, below 85 % relative humidity, dust covers fitted, packaging closed
Over 6 months	As above; also turn the conveyor belts by hand every 3 months, protect bright steel parts with corrosion inhibitor, remove the controller battery and store it separately
Before use	Carry out Section 5.4 before start-up

4 Installation

4.1 Site Requirements and Foundation

- Level concrete floor, minimum load 750 kg/m², flatness ±3 mm over the machine length
- Service clearance of 1,000 mm on the operator side and 800 mm on all other sides (Figure 4-1)
- Lighting of at least 500 lux at the operator panel and 300 lux inside the guard
- Ambient 5–40 °C; no direct sunlight on the operator panel; no condensation



Figure 4-1. Footprint and service clearances. Dots show the positions of the leveling feet.

4.2 Mechanical Installation and Leveling

1. Position sections A and B in line, using the alignment pins at the joint.
   → Pins engage; gap at the joint less than 2 mm.
2. Bolt the sections together with the M12 bolts supplied and tighten to 80 N·m (Appendix 12.8).
   → All eight bolts tight.
3. Level each section with the adjustable feet.
   → Within 0.5 mm/m in both directions, checked with a machinist's level on the frame rails.
4. Anchor the feet to the floor with the brackets supplied.
   → Feet do not move when pushed.
5. Fit the operator panel arm and stack light; plug in the panel cable.
   → Connector locked.
6. Remove all red transport locks and store them for future moves.
   → Six locks removed; gantry and swing arm move freely by hand.

4.3 Electrical Supply Connection

Item	Requirement
Supply	480 V AC ±10 %, 3-phase + PE, 60 Hz; full-load current 32 A; SCCR 65 kA
Hot-melt unit	Supplied from the machine cabinet through Q5 (included in full-load current)
Upstream protection	40 A Class J fuses or equivalent circuit breaker, by the owner
Cable	4 × 8 AWG copper, 75 °C, or per local code; entry through gland plate on cabinet top
Grounding	Separate PE conductor; resistance to building ground below 0.1 Ω

⚠ DANGER

Hazardous voltage inside electrical enclosures and terminal boxes.

Contact with live parts will cause death or severe burns.

- Only qualified electrical workers may open the enclosure.
- Switch off and lock the main disconnect before opening.
- Verify absence of voltage with a rated tester on every conductor before touching.

1. Check that the plant supply is switched off and locked.
   → Supply verified dead with a rated tester.
2. Pull the supply cable through the gland plate and tighten the gland.
   → Cable held firmly.
3. Connect L1, L2, L3 to the top of Q0 and PE to the PE bar. Tighten to the values on the terminal label.
   → Conductors tight; no bare copper visible.
4. Switch on the plant supply and Q0, then check the phase rotation by jogging the infeed in Setup mode.
   → Belt moves in the direction of the arrow. If not, swap two phases at the supply.

4.4 Compressed Air Connection

Item	Requirement
Pressure	6 bar (87 psi) at the machine, minimum 5.5 bar under load
Consumption	350 l/min (12.4 scfm) average at 20 cases/min
Quality	ISO 8573-1 class 4.4.4 (oil-free recommended)
Connection	½-inch NPT female at valve V0; hose minimum 12 mm inside diameter

NOTICE

Oil, water or dirt in the compressed air.

Contamination damages valves and vacuum generators and voids the warranty.

- Blow out the supply line before connecting.
- Install a plant air dryer if the dew point can exceed 3 °C.

4.5 Control and Network Connections

The SAMPLE-A exchanges signals with adjacent machines through terminal strip X10 and links the line emergency stop through safety terminal X11.

Terminal	Signal	Direction	Meaning
X10:1–2	Machine ready	Out	SAMPLE-A is in Idle or Execute
X10:3–4	Machine running	Out	SAMPLE-A is in Execute
X10:5–6	Downstream full	In	Next machine cannot accept cases
X10:7–8	Upstream running	In	Products can be expected
X11:1–4	Line E-stop in	In (2-channel)	Stops the SAMPLE-A if any line E-stop is pressed
X11:5–8	Machine E-stop out	Out (2-channel)	Tells the line that a SAMPLE-A E-stop is pressed
RJ45	Plant network	—	Production data, time sync, option R

4.6 Noise and Vibration Measures

Install the machine on its leveling feet without additional damping; the frame is designed to be rigid. Route exhaust air from the valve manifold to the silencer supplied. Do not place the machine against hollow walls, which amplify noise.

5 Commissioning

5.1 Pre-Start Checklist

Complete and sign this checklist before the first start. Keep the signed copy with the commissioning report.

Check	Requirement / expected result	OK
Transport locks	All six red locks removed	☐
Guards	All fixed guards bolted; doors close and lock	☐
Leveling	Within 0.5 mm/m; anchors fitted	☐
Supply voltage	480 V ±10 % measured at Q0, all three phases	☐
Grounding	PE connected; continuity below 0.1 Ω	☐
Phase rotation	Infeed runs in arrow direction	☐
Air	6 bar at regulator; no leaks audible	☐
Glue	Tank filled with specified adhesive; unit heats to 175 °C	☐
Emergency stops	Each of E1–E3 stops the machine (Section 9.4)	☐
Door interlocks	Each of D1–D4 stops the machine	☐
Light curtain	Test rod stops the erector	☐
Line interface	X10 signals and X11 line E-stop tested with adjacent machines	☐
Passwords	Default passwords changed for all user levels	☐

5.2 Initial Start-Up and Test Run

1. Switch on main disconnect Q0.
   → Operator panel starts in about 60 s and shows A103.
2. Log in at Setup level.
   → User name shown in the panel header.
3. Press **Reset**.
   → Blue light goes out; air pressure builds; gauge reads 6 bar.
4. Press **Home** on the *Manual Control* screen.
   → Gantry, fingers and swing arm move to home; A122 clears.
5. Load recipe F02 on the *Recipes* screen.
   → Recipe name shown on *Home*.
6. Turn the mode key to AUTO and run 10 dry cycles without product.
   → Machine erects 10 empty cases without alarms.
7. Run 20 cases with product at 10 cases/min.
   → All cases square, full and sealed.
8. Increase speed to 20 cases/min and run 200 cases.
   → Efficiency above 98 %; no repeated alarms.
9. Record results in the commissioning report.

5.3 Safety Function Verification

Function	Test	Pass criterion
E-stop E1–E3	Press each while running	All motion stops within 0.5 s; restart needs Reset
Doors D1–D4	Open each while running	Stops; guard locking holds D1/D2 closed until standstill
Light curtain LC1	Insert 30 mm test rod at three heights	Erector stops each time
STO	Press E-stop; check drive status	All drives show STO active
Brake	Run brake test from *Maintenance* screen	Pass shown; head does not move
Dump valve	Press E-stop	Gauge falls to 0 bar within 2 s

5.4 Re-commissioning After Standstill or Relocation

After more than 6 months out of service, or after relocation: check belts and vacuum cups for aging, drain the air filter, replace the glue, refit the controller battery, then repeat the complete Pre-Start Checklist (Section 5.1) and the safety-function verification.

5.5 Operator Training and Handover

Each operator needs a 4-hour training session covering Chapters 1, 6, 7 and 8 of this manual, including a practical emergency stop and jam-clearing exercise. Setup technicians need an extra 4 hours on changeover. Record training on the sign-off sheet. The machine is handed over when the commissioning report, the training records and this manual have been signed for by the owner.

6 Controls and Operator Panel

6.1 Operator Panel Layout

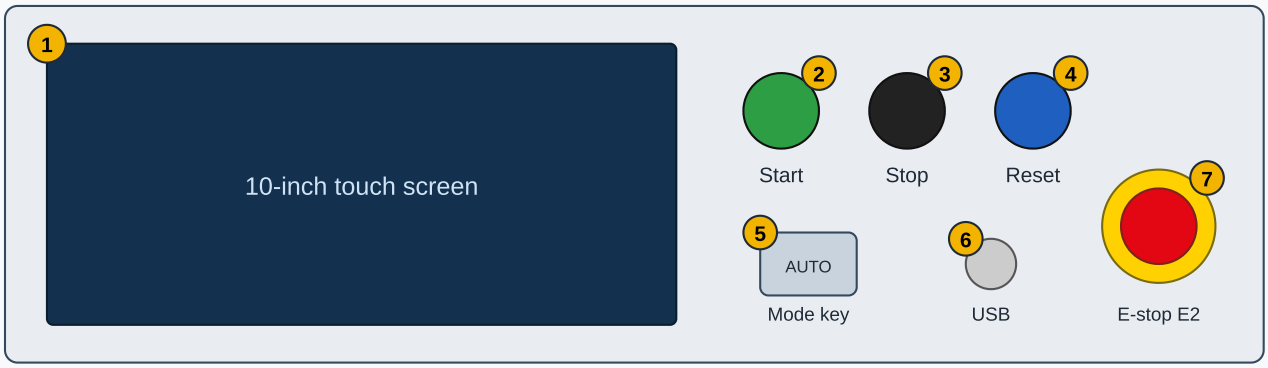


Figure 6-1. Operator panel.

No.	Element	Function
1	Touch screen	Shows status, alarms and all screens in Section 6.4
2	**Start** (green, lit)	Starts automatic production; lit while running
3	**Stop** (black)	Cycle stop at the end of the current case
4	**Reset** (blue, lit)	Resets the safety circuit and red alarms; lit when reset is needed
5	Mode key switch	AUTO / SETUP / CLEAN; key removable in AUTO only
6	USB port (covered)	Recipe backup and alarm export, Maintenance level
7	Emergency stop E2	Immediate stop of all motion

The main disconnect Q0 is on the cabinet door; emergency stops E1 and E3 are at the infeed and discharge ends.

6.2 Stack Light

Color	Steady	Flashing
Green	Running (Execute)	Starting or Suspended
Amber	Warning active, production continues	Glue heating up
Red	Stopped by fault or E-stop	Safety controller fault
Blue	Operator action needed (reset, reload, maintenance due)	—

6.3 User Levels and Login

Level	Who	Can do
Operator	All trained operators (no password)	Start, stop, reset, view alarms, select an approved recipe
Setup	Setup technicians	Plus: Setup mode, edit recipes within limits, changeover

Level	Who	Can do
Maintenance	Maintenance staff	Plus: hour counters, brake test, USB export
Engineer	Manufacturer or authorized engineer	Plus: recipe limits, network settings, remote access

Log in by touching the user name in the panel header. The panel logs out to Operator after 10 minutes without a touch.

6.4 Screen Navigation

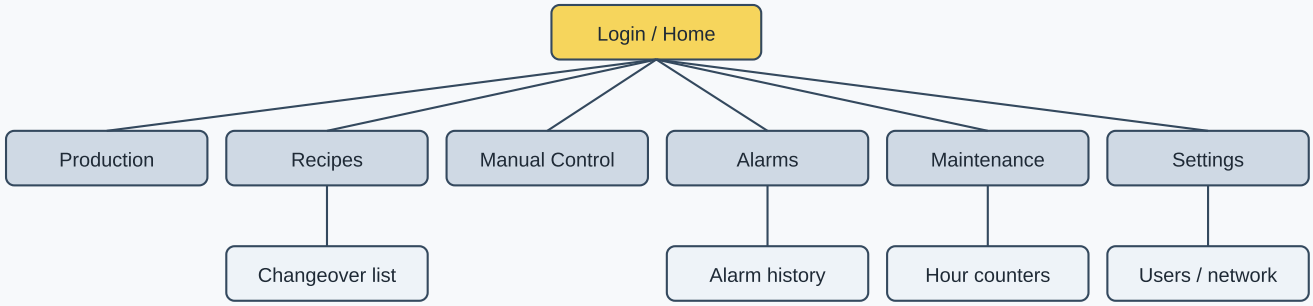


Figure 6-2. Screen map. The tab bar at the bottom of every screen leads to the main screens.

6.5 Home Screen

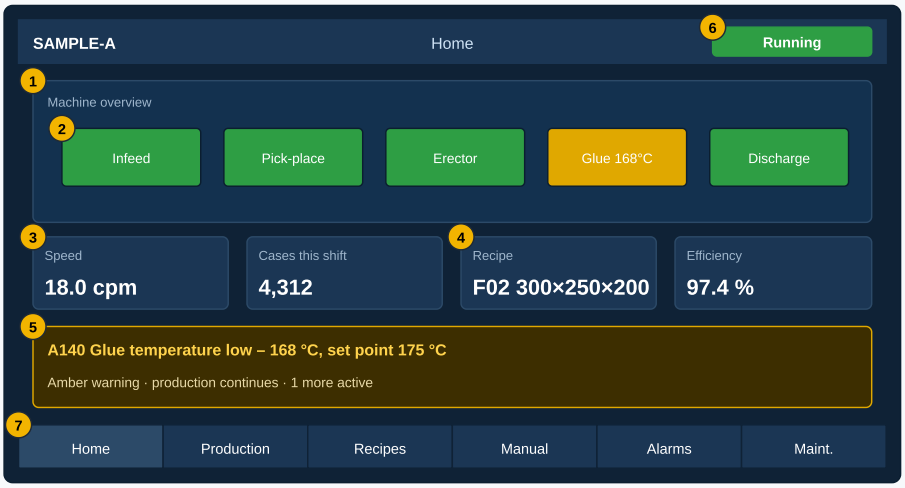


Figure 6-3. *Home* screen.

No.	Element	Function
1	Machine overview	One tile per station; green = OK, amber = warning, red = fault, grey = stopped
2	Station tile	Touch to open the station detail with sensor states
3	Speed and counters	Actual speed and good cases this shift
4	Recipe	Active recipe; touch to open *Recipes*
5	Alarm banner	Most recent active alarm; touch to open *Alarms*
6	Machine state	Current state from Section 2.12

No.	Element	Function
7	Tab bar	Main screen navigation

6.6 Recipes Screen



Figure 6-4. *Recipes* screen.

No.	Element	Function
1	Recipe list	Up to 20 stored formats
2	Recipe values	Values of the highlighted recipe; full list in Section 7.5
3	Value field	Touch to edit (Setup level); limits are shown and enforced
4	Load recipe	Loads the highlighted recipe; machine must be Stopped
5	Save	Saves changes; previous version kept for undo
6	Changeover list	Opens the step list for changing to this format

6.7 Manual Control Screen



Figure 6-5. *Manual Control* screen (Setup mode only).

No.	Element	Function
1	Axis list	Select the axis or actuator to move
2	Position	Actual position of the selected axis
3	Jog keys	Move while held, together with the enabling device
4	Set Home	Sets the reference position (Section 9.9)
5	Mode banner	Reminds that safeguarding is reduced

6.8 Alarms Screen

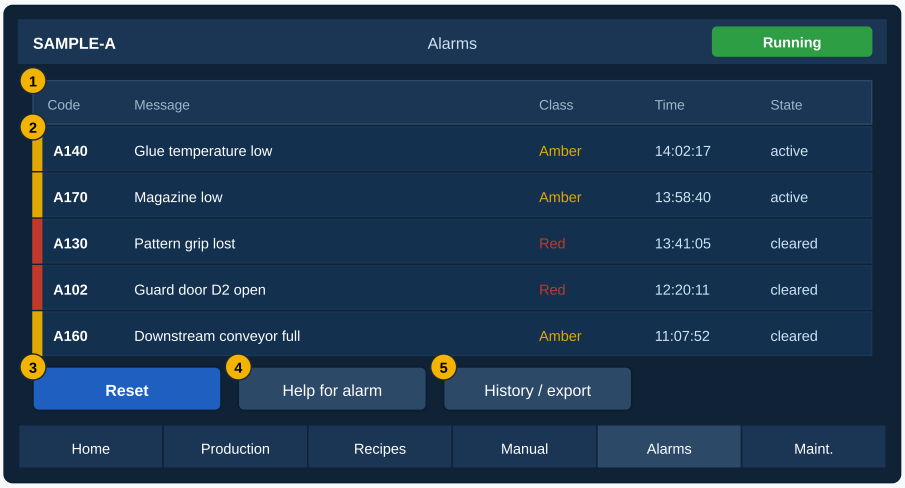


Figure 6-6. *Alarms* screen.

No.	Element	Function
1	Column header	Code, message, class, time and state

No.	Element	Function
2	Alarm row	Color bar shows class: red, amber or blue
3	Reset	Resets cleared red alarms
4	Help for alarm	Shows cause and remedy from Section 8.1
5	History / export	Last 1,000 alarms; export to USB at Maintenance level

6.9 Maintenance Screen

Shows running hours and cycle counters per station, the next due date of each task in Section 9.2, and the brake test. Confirm each completed task here to reset its reminder (A190).

7 Operation

7.1 Operating Modes

Mode	How to select	What happens
Automatic	Key switch to AUTO, all doors closed	Full-speed production; all guards active
Setup	Key switch to SETUP, Setup login	Single-axis jog at reduced speed (max. 10 mm/s) with the enabling device; door D1 may be opened after standstill
Clean	Key switch to CLEAN	Conveyors run at 10 % speed for belt cleaning while **Start** is held; all other motion locked

Changing the mode stops the machine. The mode key can be removed only in AUTO, so the person holding the key controls access to Setup and Clean.

⚠ WARNING

Safeguards partly muted in setup, teach or maintenance mode.

Moving parts can reach the operator.

- Use the enabling device; release it to stop motion.
- Only one person in the danger zone controls motion.
- Return to automatic only after everyone has left the zone.

7.2 Daily Start-Up

1. Check that all guards are closed and nobody is inside the guarded area.
   → All doors closed; area clear.
2. Open air valve V0.
   → Gauge on the filter regulator reads 6 bar.
3. Switch on main disconnect Q0.
   → Operator panel starts within 60 s; stack light blue.
4. Switch on the hot-melt unit if it was switched off.
   → Glue unit display shows heating; stack light amber flashing.
5. Wait for the glue to reach temperature.
   → Glue unit shows ready (about 20 minutes); amber light goes out.
6. Load case blanks into the magazine (Section 7.3).
   → A170 clears.
7. Press **Reset**.
   → Blue light goes out; machine re-homes; state Idle.
8. Check the active recipe on the *Home* screen; load the correct one if needed.
   → Correct format shown.
9. Press **Start**.
   → Stack light green; first case is erected within 5 s.
10. Inspect the first three cases.
    → Cases square, correct count, flaps fully closed and glued.

7.3 Normal Production

- Keep the blank magazine filled; the low-blank warning appears at 20 blanks (about 1 minute of production).
- Watch the first cases after any restart for correct glue and flap closing.

- Respond to the blue stack light (operator action needed) and to alarms promptly.
- Check the glue level every two hours or when A141 appears.

Loading case blanks

1. Take a bundle of blanks and remove the strapping outside the machine.
   → Blanks flat and dry.
2. Square the bundle by knocking it on a flat surface.
   → Edges aligned.
3. Place the bundle on the magazine rails behind the light curtain, printed side facing the arm.
   → Bundle rests on both rails; LC1 not interrupted.
4. Push the bundle forward against the pusher plate.
   → Pusher plate moves; A170 clears.

Refilling the glue tank

⚠ WARNING

Molten adhesive in the tank at up to 180 °C.

Splashes cause severe burns.

- Wear heat-resistant gloves and safety glasses.
- Open the lid from the side, never lean over the tank.

1. Open the tank lid.
   → Glue level visible.
2. Add pellets of the specified adhesive up to 20 mm below the rim. Do not overfill.
   → Level below the max mark.
3. Close the lid.
   → A141 clears within a few minutes.

7.4 Format Changeover

Changeover takes about 15 minutes with two people. The change parts for each format are color-coded. Select the new recipe first: the *Changeover list* then guides you step by step and the machine will not start until all steps are confirmed.

Change part	Where	Tool
Collation finger set	Collation station	Quick-release pins, no tools
Pick-head plate	Pick head	Two star knobs
Magazine side guides	Blank magazine	Handwheel with scale
Flap-closer width	Glue station	Handwheel with counter
Infeed side guides	Infeed	Handwheel with scale

1. Let the machine run empty and press **Stop**.
   → No products or cases in the machine; state Stopped.
2. Load the new recipe on the *Recipes* screen and open the *Changeover list*.
   → List of steps shown.
3. Turn the mode key to SETUP and lower the pick head to the change position.
   → Head at change height.

⚠ WARNING

Sharp tools, blades, knives and cut edges.

Contact can cause deep cuts or amputation.

- Handle tools and blades with cut-resistant gloves.
- Use tool holders or guards when changing blades.
- Never reach past a guard to remove scrap.

4. Open door D1 and replace the collation finger set with the color of the new format.
   → Pins locked; color matches the screen.
5. Replace the pick-head plate.
   → Plate seated on both dowels; knobs tight.
6. Set the magazine side guides to the value shown.
   → Scale reads the displayed value.
7. Set the flap-closer width with the handwheel.
   → Counter reads the displayed value.
8. Set the infeed side guides.
   → Products slide freely with 2–3 mm clearance.
9. Close door D1, turn the key to AUTO and confirm the last step on the screen.
   → Changeover list complete; A103 shown.
10. Press **Reset**, then **Start**, and run 5 test cases.
    → Test cases square, full and sealed.

7.5 Recipe Parameters

Parameter	Unit	Range	Effect
Case length / width / height	mm	200–450 / 150–350 / 100–350	Erector, gantry place height, flap closer
Pattern rows × lanes	—	1–6 × 1–6	Collation and pick-head plate
Products per case	—	1–36	Count check
Maximum speed	cases/min	5–20	Belt speeds and gantry motion times
Pick height offset	mm	–20 to +20	Fine adjustment of grip height
Place height above case bottom	mm	5–30	Release height
Glue dot length	mm	8–30	Amount of glue per dot
Glue dot pitch	mm	40–150	Spacing between dots
First dot from leading edge	mm	20–100	Glue pattern position
Compression time	s	0.8–3.0	Time under pressure

7.6 Stopping and Shutdown

Stop	Use	Restart
Cycle stop (**Stop**)	Normal end of production	Press **Start**
Hold (touch key)	Short pause, e.g. to change pallets downstream	Press **Start**
Door open	Access in Setup mode	Close door, **Reset**, **Start**
Emergency stop	Danger to people or machine	Release button, **Reset**, **Start**

End-of-shift shutdown

1. Stop the upstream supply of products and let the machine run empty.
   → No products on the infeed.
2. Press **Stop**.
   → Stack light red; all stations at home.
3. Set the hot-melt unit to standby (if production resumes within 8 hours) or switch it off.
   → Glue unit shows standby or off.
4. Clean the machine (Section 7.7).
5. Switch main disconnect Q0 to OFF.
   → Panel dark.
6. Close air valve V0.
   → Gauge reads 0 bar.

7.7 Cleaning

Clean the machine at the end of each shift: remove dust and carton scraps with a vacuum cleaner (not compressed air), and wipe the sensor lenses with a soft cloth. Clean the belts in Clean mode with a damp cloth. Clean the glue nozzles only when hot, using the tools from the glue unit kit (Section 9.6).

> **Note** Do not use solvents on belts or the operator panel. Use water with a mild detergent only.

8 Alarms and Troubleshooting

8.1 Alarm Messages

Alarms appear on the *Alarms* screen with a colored banner. **Red** alarms stop the machine; **amber** alarms are warnings that let production continue; **blue** alarms ask the operator to do something. Correct the cause, then press **Reset**. The stop reaction of each alarm is listed in Appendix 12.6.

Code	Message	Class	Probable cause	Remedy
A100	Safety controller fault	Red	Safety controller reports an internal or wiring fault	Note the diagnostic code on the *Alarms* screen; call electrical maintenance
A101	Emergency stop pressed	Red	An emergency stop button (E1–E3) is pressed	Remove the cause, pull out the button, press **Reset**
A102	Guard door open	Red	Door D1–D4 opened in Automatic; the door number is shown	Close the door, press **Reset**
A103	Safety reset required	Blue	Safety circuit is ready to be reset	Make sure nobody is inside the guard, press **Reset**
A104	Safe Torque Off feedback mismatch	Red	A drive did not confirm torque off, or wiring fault	Do not restart; call electrical maintenance to test the STO circuit (Section 9.4)
A110	Infeed drive fault	Red	Motor overload, product jam or drive trip on the infeed	Lock out, clear the jam, check belt tension (Section 9.5); reset the drive
A111	Infeed backup	Amber	Products queued at the backup sensor for more than 3 s	Check collation for a fallen product; infeed runs at half speed until clear
A112	Product count mismatch	Amber	Products counted do not match collation rows	Check the metering sensor alignment (Section 9.8); remove doubled products
A120	Gantry horizontal following error	Red	Axis blocked, belt slipping or mechanical binding	In Setup mode check for obstruction; check timing belt and coupling
A121	Gantry vertical following error	Red	Head blocked by a product or case, or load too heavy	Remove obstruction; check recipe product weight
A122	Axes not referenced	Blue	Power was cycled or an axis was replaced	Press **Reset**, then **Home**
A123	Vertical brake test failed	Red	Holding brake worn or brake supply fault	Do not work under the head; lower it to rest; call maintenance
A125	Pattern not complete	Amber	Collation did not complete a pattern within 10 s	Check for missing lanes or a stuck finger
A130	Pattern grip lost	Red	Vacuum at the pick head dropped during travel	Check pick-head cups and vacuum generator; check product surface
A135	Case blank not picked	Red	No vacuum at swing-arm cups after one retry	Check blanks for warping and cups for wear (Section 9.7)
A136	Case not square	Red	Case-square sensor not made after forming	Check blank quality and magazine guides; remove the case
A137	Flap folder timeout	Red	A flap cylinder did not reach its end position in 0.8 s	Check air pressure, valve and cylinder sensor
A140	Glue temperature low	Amber	Glue tank or hose below set point minus 10 °C	Wait for heat-up; check heater and hose connection
A141	Glue level low	Amber	Glue tank below minimum level	Refill with the specified hot-melt (Section 7.3)

Code	Message	Class	Probable cause	Remedy
A142	Glue unit fault	Red	Hot-melt unit reports heater, sensor or pump fault	See the hot-melt unit manual; check its display code
A150	Air pressure low	Red	Supply below 5 bar	Check plant air supply, valve V0 and the filter regulator
A160	Downstream conveyor full	Amber	Downstream machine not accepting cases	Clear downstream; machine resumes automatically
A161	Discharge jam	Red	Case stayed at the exit sensor for more than 4 s	Remove the case; check for open flaps
A170	Magazine low	Amber	Fewer than 20 blanks left	Reload the magazine (Section 7.3)
A171	Magazine empty	Blue	No blank detected	Reload the magazine; press **Start**
A180	Light curtain interrupted	Red	Reach into the magazine while the erector runs	Withdraw, press **Reset**
A190	Maintenance due	Blue	A preventive maintenance interval has elapsed	Carry out the task shown; confirm on *Maintenance* screen
A191	Cabinet temperature high	Amber	Cabinet above 45 °C; fan filter blocked	Clean or replace fan filters
A195	Controller battery low	Amber	Program backup battery below limit	Replace battery within 2 weeks with power ON

8.2 Troubleshooting Guide

Use this guide for problems that do not show an alarm, or when an alarm keeps returning. Work through the causes in the order given. Lock out the machine before working inside the guard (Section 1.10).

Infeed conveyor

Symptom	Probable cause	Remedy
Products fall over at the metering transfer	Gap between belts too large; product top-heavy	Adjust the transfer plate to 3 mm; reduce speed
Products not counted, A112 repeats	Sensor dirty or out of alignment	Clean lens; align sensor (Section 9.8)
Belt runs off to one side	Tracking out of adjustment; uneven tension	Adjust tension evenly (Section 9.5)
Belt slips under load, A110	Tension too low; worn drive pulley lagging	Re-tension belt; replace pulley lagging
Infeed backs up although collation is empty	Upstream sends products faster than recipe speed	Match upstream speed; check A111 history

Collation and pick-and-place

Symptom	Probable cause	Remedy
Products fall during transfer, A130	Wrong recipe; worn or dirty pick-head cups; low vacuum	Check recipe; clean or replace cups; check vacuum generator VG1 filter
Pattern placed off-center in case	Pick-head plate loose; axis reference lost	Tighten plate; re-reference axis (Section 9.9)
Gantry noisy or vibrating	Timing belt tension wrong; guide carriage dry	Check belt (Section 9.5); lubricate guides (Section 9.3)
Following error A120/A121 at high speed	Payload above recipe weight; worn coupling	Check product weight; inspect coupling
Rows incomplete, A125	Finger set does not match recipe; lane blocked	Fit correct finger set; clear lanes

Case erector

Symptom	Probable cause	Remedy
Frequent blank misfeeds, A135	Humid or warped blanks; worn cups	Store blanks dry; replace cups (Section 9.7)
Two blanks picked at once	Magazine pusher pressure too high; blanks stuck together	Reduce pusher pressure; fan the blanks before loading
Case not square, A136	Magazine guides out of adjustment; blank out of tolerance	Set guides to recipe value; check blank dimensions
Bottom flaps not fully closed	Flap cylinder slow; low air	Check air pressure; adjust flow control
Erector stops for no visible reason	Light curtain dirty or misaligned	Clean LC1 front screens; check alignment LEDs

Glue and flap closing

Symptom	Probable cause	Remedy
Cases open after closing	Glue too cold; too little glue; compression time too short	Check glue temperature (175 °C); clean nozzles; increase dot length or compression time
Glue strings or drips	Glue too hot; nozzle worn	Reduce temperature by 5 °C; replace nozzle
No glue on one line	Nozzle blocked; gun valve faulty; hose heater failed	Clean nozzle (Section 9.6); check gun and hose on glue unit display
Glue dots in wrong place	Encoder slipping; wrong recipe values	Check encoder wheel contact; check recipe values
Top flaps crushed or creased	Flap-closer width wrong	Set width to recipe value
Glue charring in tank	Temperature too high; tank not emptied for months	Reduce temperature; clean tank (Section 9.6)

Case discharge

Symptom	Probable cause	Remedy
Cases jam at exit, A161	Open flap catching on guard; rollers not turning	Check sealing; check drive chain and motor
Machine stops often with A160	Downstream machine slower than SAMPLE-A	Reduce SAMPLE-A speed or adjust downstream

Air, electrical and controls

Symptom	Probable cause	Remedy
Machine will not start, no alarm	Mode switch not in AUTO; **Reset** not pressed; recipe not loaded	Turn key to AUTO, press **Reset**, load recipe
Reset has no effect	E-stop still pressed; door not fully closed	Check each device; status shown on *Home* station tiles
A150 during production	Plant air demand peaks; filter blocked	Check plant supply; replace filter element
Operator panel blank	Panel fuse; 24 V supply fault	Check fuse F12; check 24 V LED on power supply
Panel touch inaccurate	Calibration lost	Run touch calibration from *Settings*

8.3 Clearing Jams

1. Press **Stop** or, if the jam is in progress, an emergency stop.
   → Machine stopped.
2. Identify the jam location from the station tile on the *Home* screen.

→ Station shown red.

3. For jams outside the guard (infeed, discharge), remove the product with the hook tool without reaching into the tunnel.
4. For jams inside the guard, lock out the machine (Section 1.10).
   → Isolation points locked.
5. Remove the jammed product or case and check for damage.
   → Station clear; no bent parts.
6. Return to service as described in Section 1.10 and restart (Section 7.2).
   → Machine runs normally.

8.4 Emergencies and Rescue

If a person is trapped: press the nearest emergency stop. The pick-and-place head can be moved by hand after pressing the brake release button inside door D2; support the head first, as it will drop slowly. Call first aid.

Burns from hot-melt: cool the area with cold water for at least 15 minutes. Do not pull solidified glue off the skin. Get medical help.

Glue fire: switch off the glue unit at Q5 and use a CO_2 or dry-powder extinguisher. Never use water on hot glue.

Power failure: the machine stops; the vertical brake holds the head. When power returns, the machine stays stopped until **Reset** is pressed.

9 Maintenance

9.1 Maintenance Safety

Lock out the machine before any maintenance task (Section 1.10). Only the personnel listed in Section 1.2 may carry out these tasks. After maintenance, refit all guards and test the safety devices before returning the machine to production. Use only original spare parts (Chapter 10).

⚠ DANGER

Hazardous voltage inside electrical enclosures and terminal boxes.

Contact with live parts will cause death or severe burns.

- Only qualified electrical workers may open the enclosure.
- Switch off and lock the main disconnect before opening.
- Verify absence of voltage with a rated tester on every conductor before touching.

9.2 Preventive Maintenance Schedule

Intervals are given in calendar time and running hours; whichever comes first applies. The *Maintenance* screen counts hours and shows reminder A190 when a task is due. Times are typical for one trained person.

Interval	Task	Who	Time
Daily	Safety devices: test each E-stop and door interlock	Operator	5 min
Daily	Sensor lenses: clean	Operator	5 min
Daily	Glue nozzles: check for drips and blockage	Operator	2 min
Daily	Machine: remove carton scraps and dust	Operator	10 min
Daily	Air gauge: check 6 bar	Operator	1 min
Weekly / 120 h	Vacuum cups: inspect for cracks and wear; replace if damaged (Section 9.7)	Maintenance	15 min
Weekly / 120 h	Filter regulator: drain condensate	Maintenance	2 min
Weekly / 120 h	Light curtain: test with test rod	Maintenance	5 min
Weekly / 120 h	Dump valve: check exhaust on E-stop	Maintenance	2 min
Weekly / 120 h	Glue nozzles: clean (Section 9.6)	Maintenance	20 min
Monthly / 500 h	Conveyor belts: check tension and tracking (Section 9.5)	Maintenance	20 min
Monthly / 500 h	Linear guides: lubricate (Section 9.3)	Maintenance	15 min
Monthly / 500 h	Flap-closer chain: lubricate	Maintenance	5 min
Monthly / 500 h	Vertical brake: run brake test	Maintenance	2 min
Monthly / 500 h	Sensors: check alignment and brackets (Section 9.8)	Maintenance	20 min
Monthly / 500 h	Fixed guards: check fixings	Maintenance	10 min
3-monthly / 1,000 h	Vacuum generator filters: replace	Maintenance	15 min
3-monthly / 1,000 h	Compression belts: inspect wear	Maintenance	10 min
3-monthly / 1,000 h	Cylinders: check for leaks and cushioning	Maintenance	20 min
6-monthly / 2,000 h	Pick-and-place timing belt: check tension; replace if worn	Maintenance	30 min
6-monthly / 2,000 h	Glue tank filter: replace	Maintenance	20 min

Interval	Task	Who	Time
6-monthly / 2,000 h	Glue tank: flush and clean (Section 9.6)	Maintenance	90 min
6-monthly / 2,000 h	Filter regulator element: replace	Maintenance	10 min
Yearly / 4,000 h	Safe Torque Off and complete safety circuit: functional test and record	Electrical	60 min
Yearly / 4,000 h	Electrical cabinet: check terminals; clean fan filters	Electrical	45 min
Yearly / 4,000 h	Gearboxes: check oil level and leaks	Maintenance	20 min
Yearly / 4,000 h	Safety labels: check, replace if unreadable	Maintenance	10 min
2 years	Controller battery: replace with power ON	Electrical	10 min
5 years / 20,000 h	Infeed and discharge belts: replace	Maintenance	3 h

9.3 Lubrication

Lubricate the points in Figure 9-1 at the intervals shown. Lubricant specifications are in Appendix 12.9. Wipe each grease nipple clean before greasing and wipe off surplus grease afterwards.

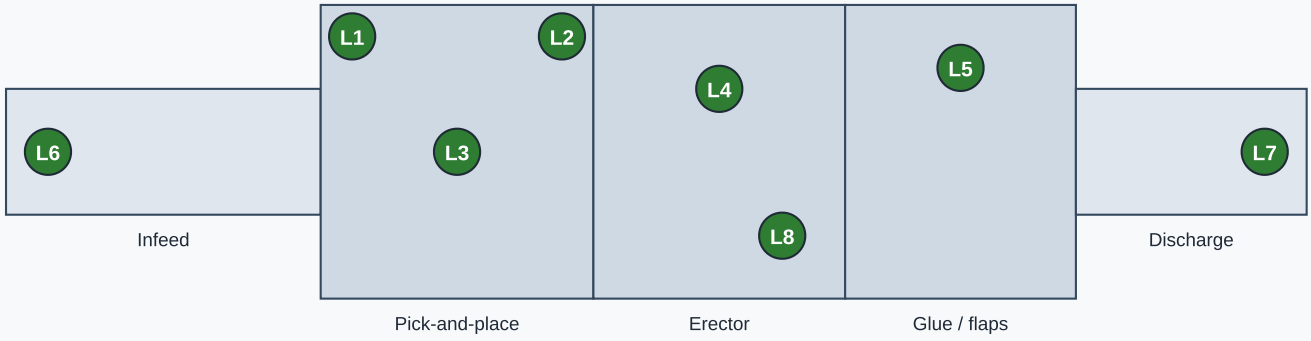


Figure 9-1. Lubrication points L1–L8.

Point	Component	Lubricant	Quantity	Interval
L1, L2	Gantry horizontal guide carriages (4)	Grease G1	2 strokes per nipple	500 h
L3	Gantry vertical guide carriages (2)	Grease G1	2 strokes per nipple	500 h
L4	Swing-arm pivot bearings (2)	Grease G1	1 stroke	500 h
L5	Flap-closer chain	Chain oil O2	Thin film	Monthly
L6	Infeed gearbox	Gear oil O1	Check level	Yearly
L7	Discharge gearbox	Gear oil O1	Check level	Yearly
L8	Magazine pusher slide	Dry PTFE spray	Light coat	Monthly

1. Lock out the machine (Section 1.10).
   → Isolation points locked.
2. Move the gantry by hand to the middle of its travel.
   → All nipples reachable.
3. Wipe each nipple clean and apply the quantity shown.
   → Fresh grease visible at the carriage seals.
4. Move the gantry by hand over its full travel three times.

→ Grease distributed; movement smooth.

5. Wipe off surplus grease and confirm the task on the *Maintenance* screen.
   → Reminder reset.

9.4 Inspection and Safety-Function Testing

Test each safety device at the interval in Section 1.6. A device passes if the machine stops and cannot be restarted until the device is reset. Record every test in the maintenance log. Take the machine out of service if any test fails.

1. Run the machine empty in Automatic.
   → State Execute.
2. Press emergency stop E1.
   → All motion stops; A101 shown; gauge falls to 0 bar.
3. Press **Start** without resetting.
   → Nothing moves.
4. Release E1 and press **Reset**, then **Start**.
   → Machine runs.
5. Repeat steps 2 to 4 for E2 and E3, and for each door D1–D4.
   → Same result for every device.
6. Insert the 30 mm test rod into the light curtain at top, middle and bottom.
   → Erector stops each time; A180 shown.
7. Record the results in the log.

9.5 Checking and Adjusting Belt Tension

Correct tension prevents slipping, tracking problems and early wear. Check tension with a spring scale at the middle of the free span (Figure 9-2).

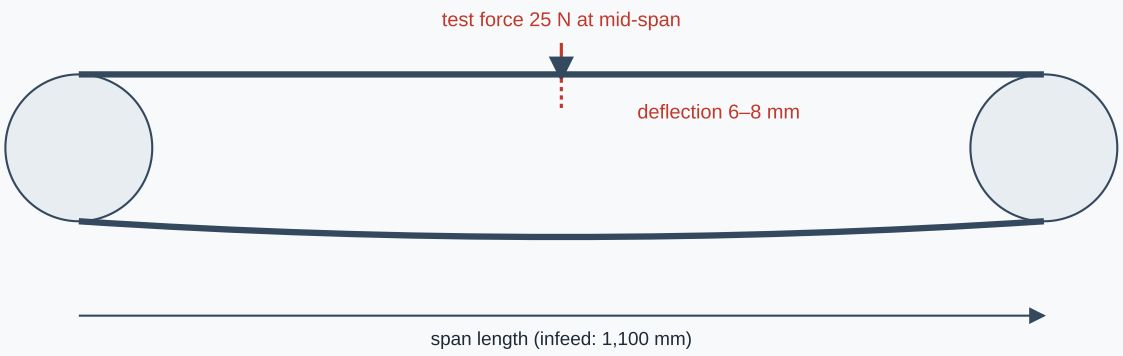


Figure 9-2. Belt tension check.

Belt	Span	Test force	Deflection
Infeed main belt	1,100 mm	25 N	6–8 mm
Discharge drive chain	600 mm	20 N	8–12 mm
Gantry timing belt (horizontal)	700 mm	30 N	3–4 mm
Compression belts	900 mm	25 N	5–7 mm

1. Lock out the machine (Section 1.10).
   → Isolation points locked.
2. Remove the side guard at the belt to be checked.

→ Belt span visible.

3. Press the belt at the middle of the span with the test force and measure the deflection.
   → Reading taken.
4. If out of range, loosen the take-up locknuts and turn both take-up screws by the same amount.
   → Deflection within the range in the table.
5. Check tracking: the belt runs centered within 2 mm on both pulleys when turned by hand.
   → Belt centered.
6. Tighten the locknuts and refit the guard.
   → Guard bolted.
7. Run the conveyor in Clean mode for 2 minutes and recheck tracking.
   → Belt stays centered.

9.6 Cleaning the Glue System

⚠ WARNING

Molten adhesive under pressure in hoses and guns.

Hot adhesive can spray out when a nozzle or hose is opened.

- Relieve pressure on the glue unit before opening any part.
- Wear heat-resistant gloves, long sleeves and safety glasses.

Nozzle cleaning (weekly)

1. Keep the glue unit at temperature; stop the machine.
   → State Stopped; glue ready.
2. Relieve glue pressure from the glue unit's service menu.
   → Pressure display reads 0.
3. Unscrew the nozzle with the nozzle wrench from the glue kit.
   → Nozzle removed.
4. Place the nozzle in the heated cleaning pot for 10 minutes, then clear the orifice with the 0.4 mm cleaning needle.
   → Light visible through the orifice.
5. Refit the nozzle with a new O-ring and tighten to 4 N·m.
   → Nozzle tight; no leak.
6. Restore pressure and fire each gun from the glue unit test function onto a scrap blank.
   → Clean, round dots from every gun.

Tank flush and filter change (every 6 months)

1. Run the tank almost empty during production.
   → Level near minimum.
2. Relieve pressure and drain the remaining glue into a metal container through the drain valve.
   → Tank empty.
3. Switch off the glue unit at Q5 and wait 30 minutes.
   → Surface below 50 °C.
4. Remove and replace the tank filter (part SA-2060).
   → New filter fitted.
5. Wipe out any charred material with a wooden scraper.
   → No black deposits.
6. Switch on, refill with fresh adhesive and wait for temperature.
   → Glue ready; A141 clear.
7. Purge each gun until clean glue flows.
   → Clear, light-colored glue.

9.7 Replacing Vacuum Cups

Worn cups are the most common cause of blank misfeeds (A135) and dropped patterns (A130). Replace all cups on a gripper at the same time.

1. Lock out the machine (Section 1.10) and close V0.
   → Gauge reads 0 bar.
2. Unscrew the worn cups from the swing arm or pick head by hand.
   → Cup holders free.
3. Check the cup holder filters and clean them if dirty.
   → Filters clear.
4. Screw in new cups (part no. SA-1104) hand-tight. Do not use tools.
   → All cups the same height.
5. Restore air and power.
   → Gauge reads 6 bar.
6. Run 5 cycles in Setup mode and check pick-up.
   → Each blank picked at the first attempt; vacuum switch on within 0.3 s.

9.8 Aligning Sensors

The machine uses retro-reflective photoelectric sensors. A misaligned sensor causes count errors, false jams or missed cases. Each sensor has an amber output LED and a green stability LED.

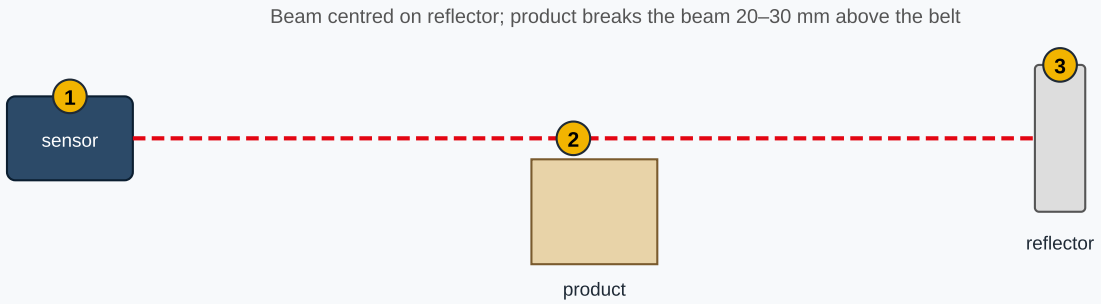


Figure 9-3. Sensor alignment: (1) sensor in slotted bracket, (2) product breaks the beam, (3) reflector.

Sensor	Function	Beam height above belt	Reflector distance
B10	Product count, metering belt	25 mm	300 mm
B11	Backup, infeed	25 mm	300 mm
B12	Row complete, collation	20 mm	450 mm
B20 / B21	Blank present / magazine low	On blank face	— (diffuse)
B23	Case square	Diagonal, 50 mm	— (diffuse)
B25	Case in loading position	100 mm	500 mm
B40	Case entry, glue station	80 mm	550 mm
B50	Case exit	80 mm	550 mm

1. Clean the sensor lens and the reflector.
   → Lenses clear.
2. Loosen the bracket screws so the sensor can be swiveled.
3. Swivel the sensor until the green stability LED is on, then find the middle of the range in which it stays on.

→ Green LED steady across the middle of the swivel range.

4. Tighten the bracket screws to 3 N·m.
   → Sensor fixed; green LED still on.
5. Pass a product through the beam.
   → Amber LED switches each time; count increases by one on the station detail screen.

9.9 Adjustment and Calibration

After replacing a servo motor, coupling or timing belt, re-reference the axis: in Setup mode, open the *Manual Control* screen, select the axis, jog it to the reference mark (engraved arrow on the rail), and press **Set Home**. Then run the brake test and 10 dry cycles.

The compression-belt encoder needs no calibration; after replacement, check that the glue pattern is in place on 5 test cases and adjust the first-dot distance in the recipe if needed.

10 Spare Parts

10.1 Recommended Spare Parts

Keep the parts marked **A** in stock at the site. Part numbers refer to the illustrated parts catalog IPC-SA.

Criticality	Meaning
A – Critical	Failure stops production and the part has a lead time; keep in stock
B – Wear	Wears in normal use; replace at the intervals in Section 9.2
C – Safety	Safety-relevant; replace only with an original part of the same specification
D – On request	Rarely needed; order when required

Item	Part no.	Description	Qty in machine	Stock	Crit.
1	SA-1104	Vacuum cup, 40 mm, nitrile	8	16	B
2	SA-1106	Vacuum cup holder with filter	8	2	B
3	SA-1110	Vacuum generator VG1/VG2, complete	2	1	A
4	SA-1210	Timing belt, gantry horizontal, HTD 8M	1	1	B
5	SA-1212	Timing belt, gantry vertical, HTD 8M	1	1	B
6	SA-1240	Servo motor, 1.5 kW, with brake	1	1	A
7	SA-1241	Servo motor, 1.0 kW	3	1	A
8	SA-1250	Linear guide carriage, size 25	6	2	D
9	SA-1305	Infeed conveyor belt, 2,350 mm	1	1	B
10	SA-1310	Gearmotor 0.75 kW, infeed	1	1	A
11	SA-1320	Compression belt, pair	1	1	B
12	SA-1420	Collation finger set, per format	1 set	1 set	D
13	SA-2010	Photoelectric sensor, retro-reflective, M18	7	2	A
14	SA-2012	Photoelectric sensor, diffuse, M18	2	1	A
15	SA-2020	Reflector, 50 mm	7	2	D

Item	Part no.	Description	Qty in machine	Stock	Crit.
16	SA-2030	Vacuum switch, adjustable	2	1	A
17	SA-2040	Pressure switch, 5 bar	1	1	A
18	SA-2050	Glue nozzle, 0.4 mm	4	4	B
19	SA-2052	Nozzle O-ring set	1	2	B
20	SA-2060	Glue tank filter element	1	2	B
21	SA-2070	Glue gun module	4	1	A
22	SA-2080	Heated hose, 2.4 m	4	1	A
23	SA-3001	Door interlock switch with coded actuator	4	1	C
24	SA-3010	Emergency stop button, complete	3	1	C
25	SA-3020	Light curtain pair, 30 mm resolution	1	—	C
26	SA-3100	Solenoid valve, 5/2, 24 V DC	6	1	A
27	SA-3105	Soft-start and dump valve	1	1	C
28	SA-3110	Cylinder, flap folder, 32 × 100	4	1	B
29	SA-3200	Filter regulator element, 5 µm	1	2	B
30	SA-4010	Variable-speed drive, 1.5 kW	3	1	A
31	SA-4020	Encoder, compression belt	1	1	A
32	SA-4030	Controller backup battery	1	1	B
33	SA-4040	Cabinet fan filter mat	2	4	B
34	SA-0090	Head support bar (lockout tool)	1	—	C
35	SA-9001	Safety label set	1	1	C

Note Parts with criticality **C** must be replaced only with original parts of the same specification. After replacing any of them, test the safety function (Section 9.4).

10.2 Ordering Parts and Service

Order parts from Example Machinery Co. Customer Service (fictional): parts@example-machinery.example, +1 555 0100. Always quote the machine model, serial number, part number and quantity. Critical parts (A) are shipped the same day if ordered before 2 PM.

11 Decommissioning and Disposal

11.1 Decommissioning and Dismantling

1. Run out all products and remove remaining case blanks.
   → Machine empty.
2. Drain the glue tank while hot into a metal container.
   → Tank empty.
3. Back up recipes and alarm history to USB.
   → Files saved.

4. Lock out all energy sources (Section 1.10).
   → Isolation points locked.
5. Disconnect power, air and network connections at the supply side.
   → Machine free of all utilities.
6. Fit the transport locks and separate the two sections.
   → Six locks fitted; joint bolts removed.
7. Lift and move the sections as described in Section 3.2.

11.2 Disposal

Material	Approx. share	Disposal
Steel and stainless steel	85 %	Metal recycling
Aluminum (profiles, gantry)	6 %	Metal recycling
Plastics (guards, belts, cables)	4 %	Plastic recycling or waste to energy
Electrical and electronic components	3 %	Electrical waste (WEEE or local equivalent)
Controller battery	—	Battery collection; remove before disposal
Lubricants, adhesive residues	—	Hazardous waste

12 Appendices

12.1 Technical Data

Parameter	Value
Model	SAMPLE-A hot-melt case packer, top-load, regular slotted case (RSC)
Output	5–20 cases/min, depending on format and pattern
Case size range (L × W × H)	200–450 × 150–350 × 100–350 mm
Products per case	1–36 (up to 6 rows × 6 lanes)
Maximum pattern mass	12 kg
Blank magazine capacity	150 blanks (about 8 minutes at 20 cases/min)
Overall dimensions (L × W × H)	5,300 × 1,600 × 2,100 mm
Infeed / discharge height	900 ± 50 mm (adjustable feet)
Machine mass	2,600 kg in two sections (1,150 kg and 1,450 kg)
Electrical supply	480 V AC ±10 %, 3-phase + PE, 60 Hz
Full-load current / SCCR	32 A / 65 kA
Control voltage	24 V DC
Drives	3 variable-speed drives (infeed, compression belts, discharge); 4-axis servo drive (gantry horizontal, gantry vertical, collation, swing arm)
Controller	Compact controller with integrated safety controller, industrial Ethernet fieldbus
Operator panel	10-inch color touch screen
Compressed air	6 bar (87 psi), 350 l/min, ISO 8573-1 class 4.4.4
Hot-melt unit	10 kg tank, 4 guns, 160–180 °C, EVA adhesive
Noise emission	74 dB(A) at the operator position (Section 1.12)
Ambient conditions	5–40 °C, 10–85 % RH non-condensing, indoor use, up to 1,000 m altitude
Safety performance	Emergency stop and guard interlocks PL d, Category 3 (ISO 13849-1)

12.2 Electrical Overview

Figure 12-1 shows the power and control distribution in simplified form. It is not a wiring diagram; always work from electrical schematic E-SA kept in the cabinet pocket.

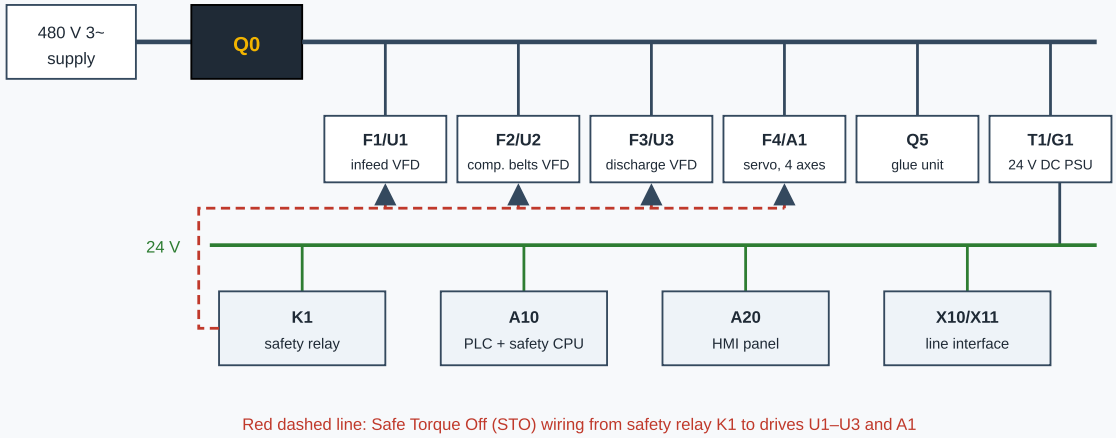
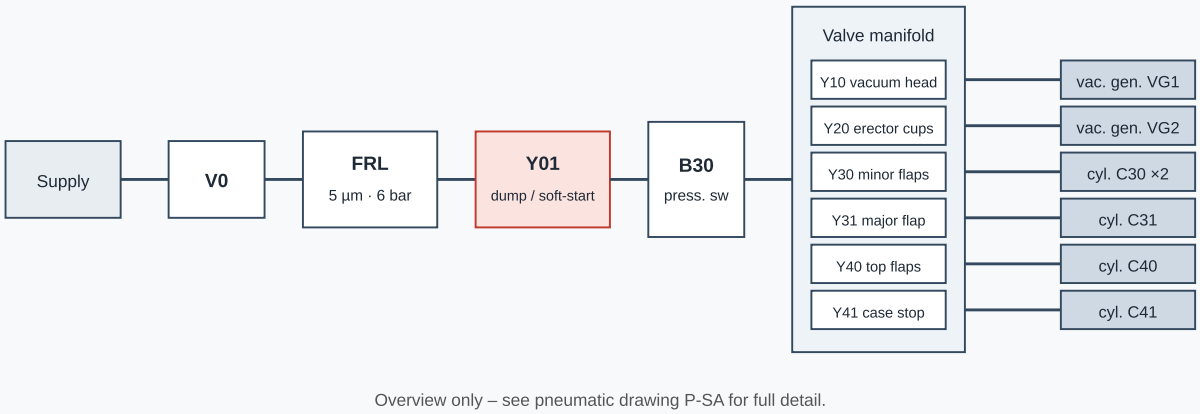


Figure 12-1. Electrical overview (simplified).

Designation	Device	Location
Q0	Main disconnect, lockable	Cabinet door
F1–F4	Drive branch fuses	Cabinet, top rail
U1 / U2 / U3	Variable-speed drives: infeed, compression belts, discharge	Cabinet
A1	Servo drive, 4 axes, with Safe Torque Off	Cabinet
Q5	Hot-melt unit disconnect	Cabinet, side
T1 / G1	Control transformer / 24 V DC power supply	Cabinet
K1	Safety relay (STO and dump valve)	Cabinet
A10	Controller with integrated safety controller	Cabinet
A20	Operator panel	Operator station
X10 / X11	Upstream / downstream line interface terminals	Cabinet, bottom

12.3 Pneumatic Overview

Figure 12-2 shows the air supply in simplified form. Full detail is in pneumatic drawing P-SA.



Overview only – see pneumatic drawing P-SA for full detail.

Figure 12-2. Pneumatic overview (simplified).

Designation	Device	Function
V0	Lockable ball valve	Isolation of the air supply (lockout point)
FR1	Filter regulator, 5 µm, with gauge	Sets 6 bar; drains condensate
Y01	Soft-start and dump valve	Builds pressure gradually; exhausts on every stop
B30	Pressure switch, 5 bar	Air pressure OK signal
VG1 / VG2	Vacuum generators	Pick head / swing arm
C30–C41	Cylinders	Minor and major flap folders, magazine pusher, flap tucker

12.4 I/O List

Hardware inputs and outputs of controller A10. The tag column gives the name used in the control program; it is provided for electrical maintenance and is not needed to operate the machine.

Address	Signal	Device	Tag
I0.0	E-stop circuit E1–E3 OK (channel A)	Safety relay K1	SI_EStopA
I0.1	E-stop circuit E1–E3 OK (channel B)	Safety relay K1	SI_EStopB
I0.2	Doors D1–D4 closed and locked	Interlocks D1–D4	SI_DoorsOK
I0.3	Light curtain clear	LC1	SI_LC1Clear
I0.4	STO feedback, all drives	U1–U3, A1	SI_STOFbk
I0.5	Start button	Operator panel	PB_Start
I0.6	Stop button	Operator panel	PB_Stop
I0.7	Reset button	Operator panel	PB_Reset
I1.0	Mode key AUTO	Key switch S1	SS_Auto
I1.1	Mode key SETUP	Key switch S1	SS_Setup
I1.2	Mode key CLEAN	Key switch S1	SS_Clean

Address	Signal	Device	Tag
I1.3	Enabling device	Pendant S2	SS_Enable
I1.4	Product count, metering belt	B10	PE_Count
I1.5	Infeed backup	B11	PE_Backup
I1.6	Row complete, collation	B12	PE_RowFull
I1.7	Blank present	B20	PE_Blank
I2.0	Magazine low	B21	PE_MagLow
I2.1	Vacuum OK, swing arm	B22	VS_Arm
I2.2	Case square	B23	PE_Square
I2.3	Case in loading position	B25	PE_CaseLoad
I2.4	Air pressure OK	B30	PS_Air
I2.5	Vacuum OK, pick head	B31	VS_Head
I2.6	Case entry, glue station	B40	PE_GlueEntry
I2.7	Case exit	B50	PE_Exit
I3.0	Minor flap folder extended	C30 sensor	LS_MinorExt
I3.1	Major flap folder extended	C31 sensor	LS_MajorExt
I3.2	Glue unit ready	Hot-melt unit	GL_Ready
I3.3	Glue level low	Hot-melt unit	GL_LevelLow
I3.4	Glue unit fault	Hot-melt unit	GL_Fault
I3.5	Downstream ready	X11	LN_DownReady
I3.6	Upstream product available	X10	LN_UpAvail
I3.7	Cabinet temperature high	Thermostat B60	TS_Cabinet
Q0.0	Soft-start / dump valve	Y01	SOL_SoftStart
Q0.1	Vacuum generator, pick head	Y20	SOL_VacHead
Q0.2	Vacuum generator, swing arm	Y21	SOL_VacArm
Q0.3	Magazine pusher	Y22	SOL_Pusher
Q0.4	Minor flap folder	Y30	SOL_Minor
Q0.5	Major flap folder	Y31	SOL_Major
Q0.6	Glue guns 1–4 enable	Hot-melt unit	GL_GunEn
Q0.7	Reset button lamp	Operator panel	LT_Reset
Q1.0	Stack light green	H1	LT_Green
Q1.1	Stack light amber	H2	LT_Amber
Q1.2	Stack light red	H3	LT_Red
Q1.3	Stack light blue	H4	LT_Blue

Drives U1–U3 and servo drive A1 are controlled over the industrial Ethernet fieldbus and do not use hardware I/O except the Safe Torque Off circuit.

12.5 Control Program Tag List

Tag	Meaning	Data type / range
MachineState	Current machine state (Section 2.12)	Integer, 0–17
ModeAuto / ModeSetup / ModeClean	Selected operating mode	Boolean
ReadyToStart	All start permissives met	Boolean
AllHomed	All axes referenced	Boolean
SafetyResetOK	Safety controller reset complete	Boolean
RecipeActive	Number of the loaded recipe	Integer, 1–20
Rcp.CaseL / CaseW / CaseH	Case dimensions of the loaded recipe	Real, mm
Rcp.Rows / Lanes	Pattern of the loaded recipe	Integer
Rcp.SpeedMax	Maximum speed	Real, cases/min
Rcp.GlueDotLen / DotPitch / FirstDot	Glue pattern values	Real, mm
Rcp.CompressTime	Compression time	Real, s
ProdCount	Products counted toward the current pattern	Integer
PatternReady	Pattern complete in collation	Boolean
PickStep	Step number of the pick-and-place sequence	Integer, 0–12
ErectStep	Step number of the case erector sequence	Integer, 0–10
CaseOpen	Open case in loading position	Boolean
GlueEncPos	Compression-belt encoder position	Real, mm
GoodCases / ShiftCases	Good cases total / this shift	Double integer
AirOK	Air pressure above 5 bar for 1 s	Boolean
DownstreamFull	Downstream not ready	Boolean
AlarmWord[0..3]	Packed alarm bits (Appendix 12.6)	Double integer array
RedAlarmActive	At least one red alarm latched	Boolean
PMHours[0..7]	Running hours per station	Real, h
BrakeTestOK	Vertical brake test passed at last homing	Boolean

12.6 Alarm List with Stop Reactions

Code	Message	Class	Stop reaction	Tag
A100	Safety controller fault	Red	Cat. 0 stop	ALM_SafetyCtrl
A101	Emergency stop pressed	Red	Cat. 1 stop	ALM_EStop
A102	Guard door open	Red	Cat. 1 stop	ALM_DoorOpen
A103	Safety reset required	Blue	—	ALM_ResetReq
A104	Safe Torque Off feedback mismatch	Red	Cat. 0 stop	ALM_STOFbk
A110	Infeed drive fault	Red	Cat. 1 stop	ALM_InfeedDrv
A111	Infeed backup	Amber	None	ALM_InfeedBackup
A112	Product count mismatch	Amber	Cycle stop	ALM_CountErr

Code	Message	Class	Stop reaction	Tag
A120	Gantry horizontal following error	Red	Cat. 1 stop	ALM_XFollow
A121	Gantry vertical following error	Red	Cat. 1 stop	ALM_ZFollow
A122	Axes not referenced	Blue	—	ALM_NotHomed
A123	Vertical brake test failed	Red	Cat. 0 stop	ALM_BrakeTest
A125	Pattern not complete	Amber	Cycle stop	ALM_PatternTO
A130	Pattern grip lost	Red	Cat. 1 stop	ALM_GripLost
A135	Case blank not picked	Red	Cycle stop	ALM_BlankPick
A136	Case not square	Red	Cycle stop	ALM_NotSquare
A137	Flap folder timeout	Red	Cat. 1 stop	ALM_FlapTO
A140	Glue temperature low	Amber	Holds cases	ALM_GlueTemp
A141	Glue level low	Amber	None	ALM_GlueLevel
A142	Glue unit fault	Red	Cycle stop	ALM_GlueFault
A150	Air pressure low	Red	Cycle stop	ALM_AirLow
A160	Downstream conveyor full	Amber	Suspend	ALM_DownFull
A161	Discharge jam	Red	Cycle stop	ALM_DischJam
A170	Magazine low	Amber	None	ALM_MagLow
A171	Magazine empty	Blue	Cycle stop	ALM_MagEmpty
A180	Light curtain interrupted	Red	Cat. 1 stop (erector)	ALM_LCBreak
A190	Maintenance due	Blue	—	ALM_PMDue
A191	Cabinet temperature high	Amber	None	ALM_CabTemp
A195	Controller battery low	Amber	None	ALM_Battery

12.7 Control Program Cross-Reference

Every rung of the control program is listed below against the machine function it serves and the section of this manual that describes it. Routine names and rung numbers are given for electrical maintenance only. Total: 200 rungs in 11 routines.

Routine	Rungs	Count	Function	Described in
MainSequence	0–3	4	Mode selection	2.13, 2.12
MainSequence	4–11	8	Machine state model	2.13, 2.12
MainSequence	12–15	4	Start permissives	2.13, 2.12
MainSequence	16–19	4	Homing after reset	2.13, 2.12
MainSequence	20–23	4	Stack light and counters	2.13, 2.12
SafetyMonitor	0–4	5	Safety device status	2.13, 1.6
SafetyMonitor	5–8	4	Safe Torque Off feedback	2.13, 1.6
SafetyMonitor	9–11	3	Reset request	2.13, 1.6
SafetyMonitor	12–13	2	Diagnostic copy for remote service (option R) (housekeeping)	—
Infeed	0–5	6	Running the infeed	2.13, 2.4

Routine	Rungs	Count	Function	Described in
Infeed	6–9	4	Counting products	2.13, 2.4
Infeed	10–13	4	Backup protection	2.13, 2.4
PickPlace	0–7	8	Building the pattern	2.13, 2.5
PickPlace	8–21	14	Pick, travel and place	2.13, 2.5
PickPlace	22–24	3	Grip-loss recovery	2.13, 2.5
PickPlace	25–27	3	Brake and following-error monitoring	2.13, 2.5
CaseErector	0–8	9	Picking a blank	2.13, 2.6
CaseErector	9–17	9	Forming and bottom-flap folding	2.13, 2.6
CaseErector	18–21	4	Magazine and light curtain	2.13, 2.6
GlueFlap	0–3	4	Glue readiness	2.13, 2.7
GlueFlap	4–15	12	Glue pattern timing	2.13, 2.7
GlueFlap	16–19	4	Compression and release	2.13, 2.7
Discharge	0–5	6	Discharge conveyor control	2.13, 2.8
Discharge	6–9	4	Downstream handshake	2.13, 2.8
AirVacuum	0–3	4	Air pressure monitoring	2.13, 2.9
AirVacuum	4–7	4	Soft-start and dump valve	2.13, 2.9
Recipes	0–7	8	Recipe load and validation	2.13, 7.5
Recipes	8–12	5	Changeover checklist	2.13, 7.5
Recipes	13–15	3	Speed scaling	2.13, 7.5
Alarms	0–27	28	Alarm detection	2.13, 8.1
Alarms	28–31	4	Alarm classes and reset	2.13, 8.1
Alarms	32–33	2	Alarm history buffer and spare rung (housekeeping)	—
Maintenance	0–7	8	Hour counters and reminders	2.13, 9.2
Maintenance	8–9	2	Spare rungs reserved for options (housekeeping)	—
Total		**200**	**200 of 200 rungs accounted for**	

12.8 Tightening Torques

Thread	Property class 8.8 (N·m)	Stainless A2-70 (N·m)	Typical use
M4	3.0	2.2	Sensor brackets, small covers
M5	6.0	4.5	Guard panels, cable ducts
M6	10	7.5	Change parts, pick-head plate
M8	25	18	Guide rails, cylinder mounts
M10	49	35	Gantry carriages, motor mounts
M12	85	60	Section joint bolts, frame
M16	210	150	Floor anchors

Exceptions: glue nozzles 4 N·m; sensor bracket screws 3 N·m; servo motor flange bolts M8 at 22 N·m with medium-strength thread locker.

12.9 Lubricants and Consumables

Code	Type	Specification	Used at
G1	Grease	NLGI 2 lithium complex; food-grade NSF H1 where required	Linear guides, pivot bearings (L1–L4)
O1	Gear oil	ISO VG 220 synthetic PAO	Gearboxes (L6, L7)
O2	Chain oil	Light chain oil, food-grade NSF H1	Flap-closer chain (L5)
S1	Dry lubricant	PTFE spray	Magazine slide (L8)
HM	Hot-melt adhesive	EVA, application 160–180 °C, open time 6–10 s	Glue tank
C1	Cleaner	Mild detergent in water	Belts, panel, guards

12.10 Maintenance Log

Copy this page and keep the completed log with the machine. Record every preventive maintenance task, repair and safety-function test.

Date	Hours	Task / repair	Parts used	Safety test OK	Name / signature
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	
				☐	

12.11 Warranty and Service Contacts

The machine carries a 12-month warranty from the date of commissioning, subject to the terms of sale. The warranty does not cover wear parts (criticality B), damage from use outside the intended use (Section 1.3), or modifications not approved in writing by the manufacturer.

Service	Contact (fictional)
Technical service, 24 hours	service@example-machinery.example · +1 555 0199
Spare parts	parts@example-machinery.example · +1 555 0100
Training	training@example-machinery.example
Remote diagnostics (option R)	Request a session through technical service

12.12 Glossary

Term	Meaning
Blank	Flat, pre-cut and creased corrugated board that is formed into a case
Case-square sensor	Sensor that confirms the formed case is rectangular before loading
Collation	Grouping products into the rows and lanes of the case pattern
Compression belts	Side belts that hold the case closed while the glue sets
E-stop	Emergency stop
Enabling device	Three-position hand switch that allows jog motion in Setup mode only while held in the middle position
FLA	Full-load amperes
Following error	Difference between commanded and actual position of a servo axis
Format	A combination of case size and product pattern; stored as a recipe
Gantry	Two-axis frame that moves the pick head horizontally and vertically
HMI	Human–machine interface: the operator panel
LOTO	Lockout/tagout: isolating and locking energy sources before work
Packaging state model	Industry-standard model of packaging machine states (ISA-TR88.00.02)
PL	Performance level of a safety function according to ISO 13849-1 (a to e)
Recipe	Stored set of values for one format
RSC	Regular slotted case
Rung	One line of the control program; rung numbers appear only in Appendix 12.7
SCCR	Short-circuit current rating
STO	Safe Torque Off: drive safety function that removes motor torque
Stop category	Type of stop according to IEC 60204-1: Cat. 0 (immediate power removal) or Cat. 1 (controlled stop, then power removal)

12.13 Detailed Revision History

Revision	Chapter	Change
A	All	First release
B	8	Added alarm A141 (glue level low)
B	9	Added 2,000-hour tasks (timing belt, glue tank filter)
C	6	New operator panel screens and figures
C	10	Updated spare parts list; added criticality classes
D	1	Expanded safety chapter: residual risks by zone, safety devices with test intervals, isolation table, PPE matrix
D	2	Added theory of operation for every station, cycle timing and machine state model
D	3–5	New chapters: transport, installation and commissioning
D	7–9	Procedures rewritten as numbered steps with expected results; troubleshooting guide expanded; new maintenance procedures
D	11–12	New decommissioning chapter; I/O list, tag list, cross-reference, torque and lubricant appendices

12.14 Index

Adjustment and calibration 48
Air supply and vacuum, how it works 18
Air supply, connecting 27
Alarm list 39
Alarm stop reactions 55
Belt tension 45
Blanks, loading 35
Case discharge, how it works 17
Case erector, how it works 16
Changeover 36
Cleaning 38
Collation and pick-and-place 14
Commissioning, first start 28
Control logic description 21
Cross-reference, control program 56
Cybersecurity 12
Decommissioning 49
Disposal 50
Electrical connection 26
Electrical overview 52
Emergencies and rescue 42
Emergency stop 7
Glue refill 35
Glue station, how it works 16
Glue system cleaning 46
Home screen 31
I/O list 53
Infeed, how it works 14
Inspection and safety tests 45
Installation, mechanical 26
Intended use 5
Jams, clearing 41
Labels, safety 8
Lifting 24
Lockout/tagout 8
Lubricants 58
Lubrication 44
Machine states 20
Maintenance log 58
Maintenance schedule 43
Manual Control screen 33
Noise 11
Operating modes 35
Operator panel 30
Personal protective equipment 8
Pneumatic overview 53
Recipe parameters 37
Residual risks 10
Safety devices 7
Sensors, aligning 47
Sequence of operation 19
Spare parts 48
Stack light 30
Start-up, daily 35
Stopping and shutdown 37
Storage 25
Technical data 51
Tightening torques 57
Training 29
Transport 24
Troubleshooting 40
User levels 30
Vacuum cups, replacing 47